

Ship Detection on Marine Radar Based on YOLOv5 with Integrated YOSMR Modules

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Outline

- Current Approach
- Dataset & Preprocessing
- Experiments & Results
- Kuartis Dataset's Results

The Challenge of Ship Detection in Radar Images

- In nearshore areas, ships look similar to reefs or land on radar [1].
- Radar images include a lot of clutter like waves and shore echoes [1].
- Small ships are hard to see — they may appear weak or blurry [1].

Current Approach

- Step 1 — Raw radar → binary mask
- Step 2 — Morphological operations
 - Opening
 - Closing
- Step 3 — Connected components
- Step 4 — Clustering
- Step 5 — Land Filtering
- Step 6 — Tracking



Fig 1: Opening removes small clutter/noise, while Closing fills gaps and smooths ship shapes — together they clean the radar mask for reliable detection [2].

Problems

- **False Positives** from clutter, noise, ghost signals
- **False Negatives** on small or low-reflection vessels
- Every false positive → **unnecessary track initiation**
- Low **precision**, unreliable **recall**
- Rigid, rule-based logic; **no learning or adaptation**

Evolution of Ship Detection: Traditional → CNN → YOSMR

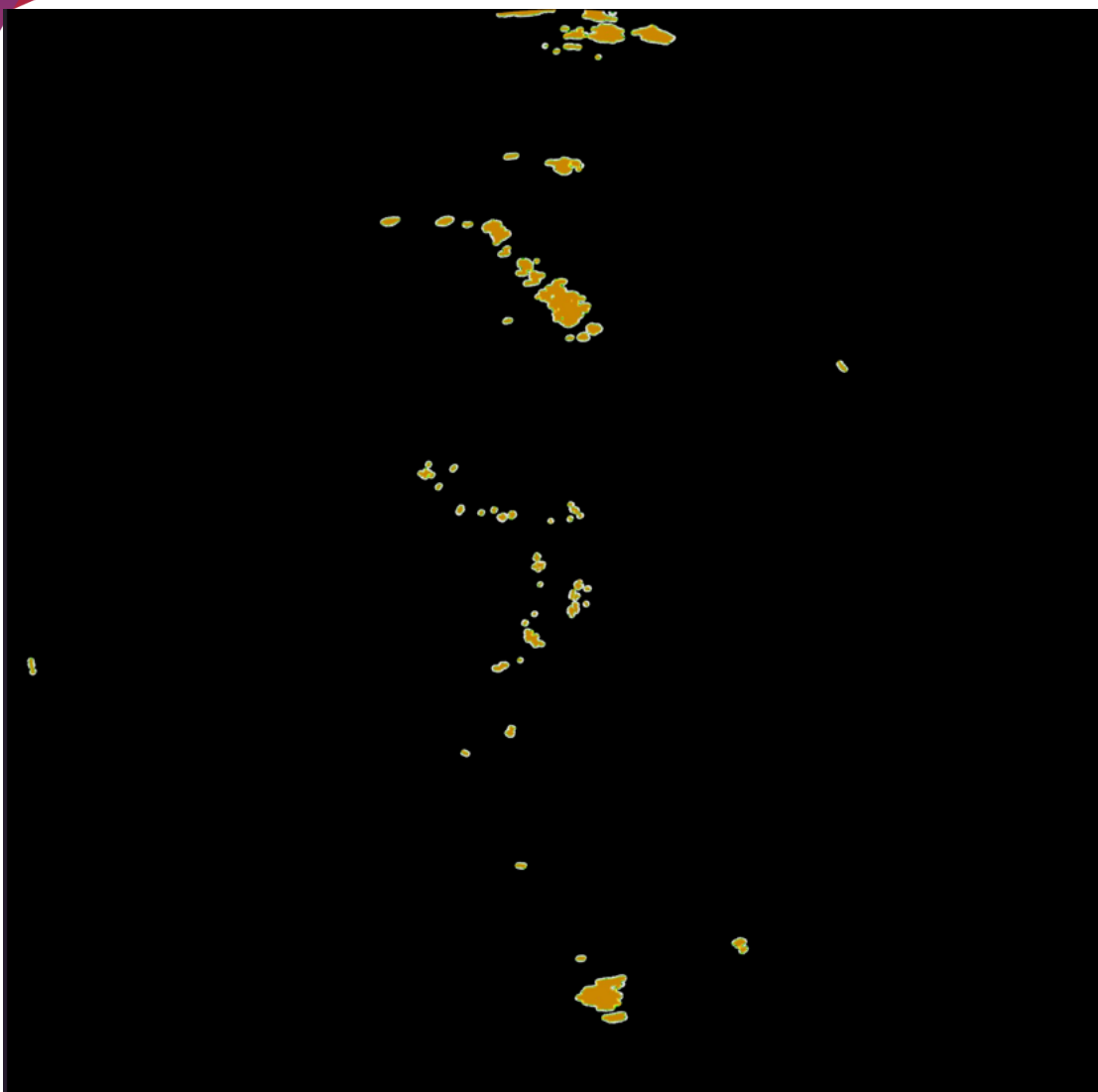
- Traditional methods → simple rules and thresholds
- CNN → learning complex patterns [3].
- YOSMR [1].
 - optimized for radar images
 - preserving small-ship features (MobileNetV3, SPP, LightPANet)
 - reducing false positives (Cluster-NMS)
 - improving localization (α -DlIoU loss)

What Makes YOSMR Different?

- Feature Extraction [1].
 - MobileNetV3 (Large)
- Feature Fusion [1].
 - LighPANet
- SPP Module [1].
- Cluster-NMS [1].
- α -DIoU Loss [1].

Dataset – MOANA X-Band Radar

- Source: MOANA maritime dataset [4].
 - X-band radar
- Format: Cartesian coordinate PNG images [4].
- Annotations: JSON with bounding boxes (*xmin, ymin, width, height*) [4].
- Class: Single class = Ship [4].



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  "copyright": "seadronix corp.",
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    "width": 2048,
    "height": 2048
  },
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      "category_id": 1,
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      "width": 23,
      "height": 15
    },
    {
      "id": 109,
      "category_id": 1,
      "xmin": 917,
      "ymin": 1320,
      "width": 22,
      "height": 25
    }
  ]
}
```

Fig 2: Each radar frame (left) is represented as a Cartesian PNG with ships highlighted, and its annotations (right) are stored in JSON format with bounding box coordinates, providing structured input for training [4].

Data & Preprocessing

1) Convert to Grayscale

- **RGB → 3-channel grayscale**

2) Convert Labels (JSON → YOLO TXT)

- JSON (xmin, ymin, w, h) → YOLO TXT (**class, x_center, y_center, w, h**)
- Class = 0 (ship)

3) Dataset Split

- 70% train | 10% val | 20% test

Dataset Split Ratios (Total: 1655 images)

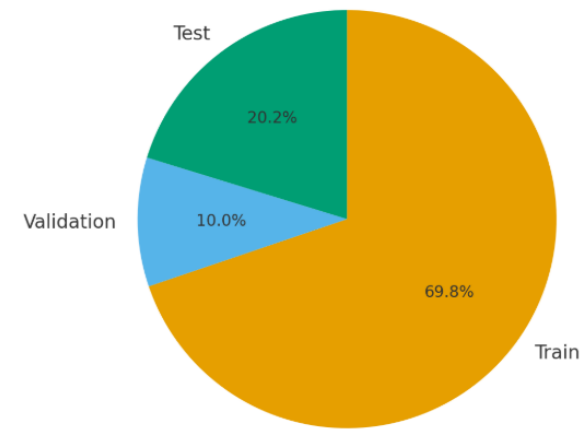


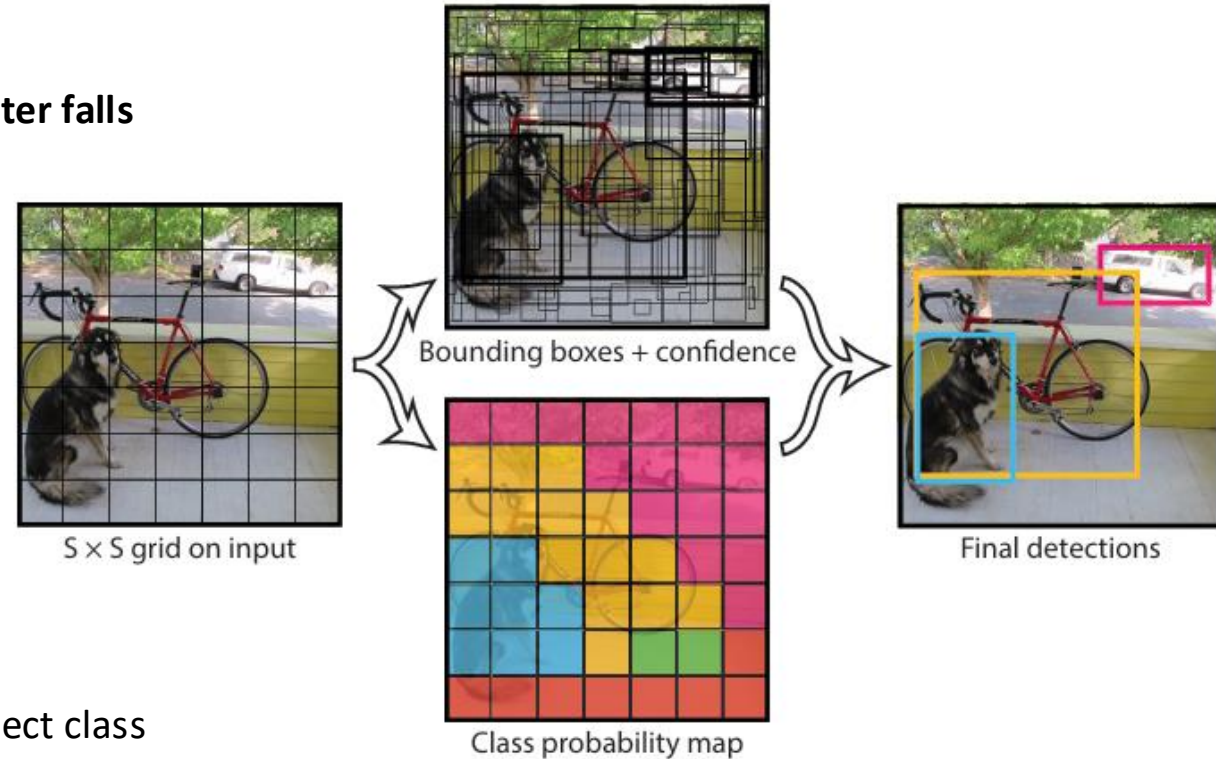
Fig 3: Dataset split into 70% training, 10% validation, and 20% testing for balanced model training and evaluation.

Training Configuration

- Optimizer: SGD (lr=0.005, momentum=0.937, weight_decay=0.0005)
- Epochs: 150 (early stopping, patience = 20)
- Batch Size: 32
- Loss Balance: box=0.05, obj=1.2, cls=0.5

YOLO Basics

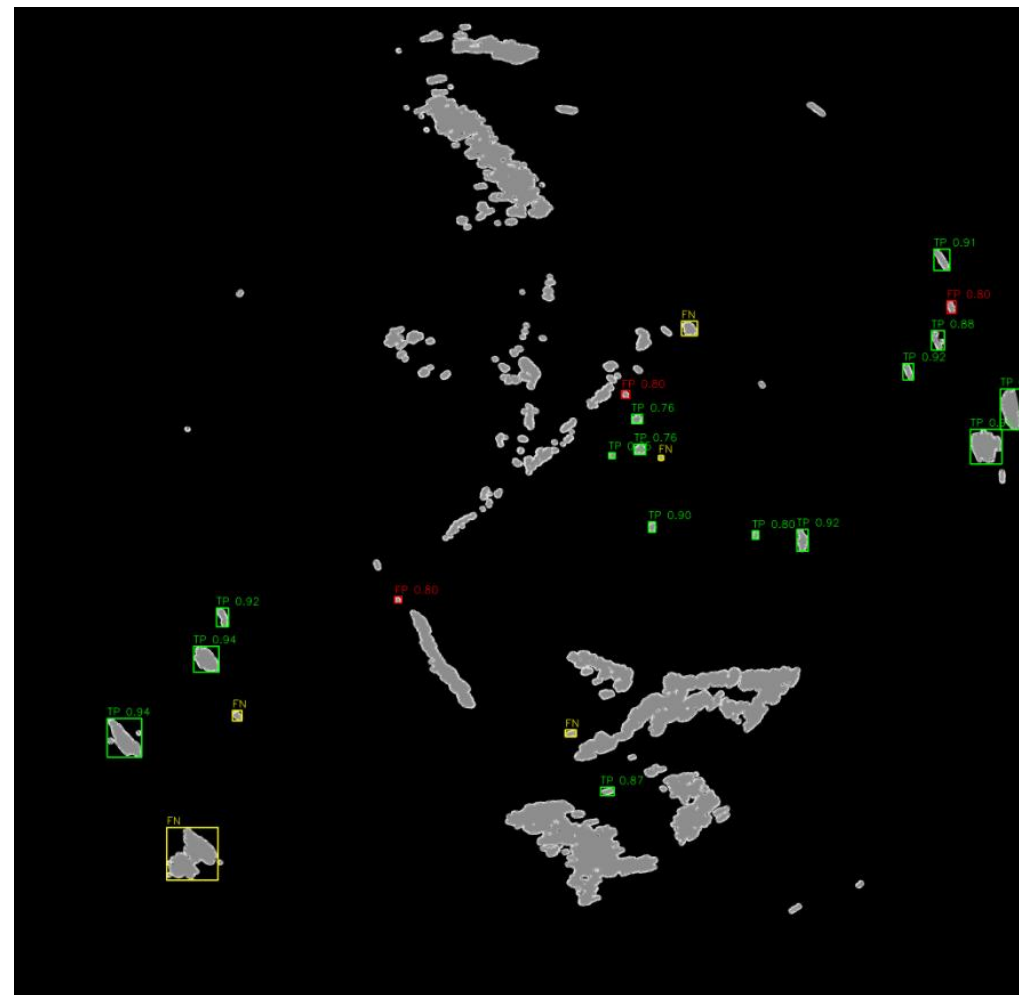
1. The image is divided into an $S \times S$ grid.
 - Each grid cell is responsible for detecting objects whose **center falls inside** that cell.
2. Each grid cell produces two things:
 - A set of **bounding boxes** centered within the grid cell, each with an **associated confidence score**.
 - How sure the model is that an object is present + how well the box fits
 - $P(\text{object exists}) \times P(\text{class} \mid \text{object exist})$
 - A **class probability map** indicating the likelihood of each object class being present in that cell.
3. Finally, this information is **combined** to produce the final **object detection results**.



Baseline

- **Model:** YOLOv5-Large
 - Balanced performance → real-time speed with strong accuracy [5]
 - Modular & extendable → easy to integrate YOSMR modules [5]

Precision	Recall	F1-Score	mAP@50	map@50:0.95
0.764	0.697	0.729	0.779	0.63



Baseline + ClusterNMS

- Instead of removing extra boxes, it merges them intelligently [1].
- A box with higher confidence has more weight in deciding the final box position and size [1].

	Precision	Recall	F1-Score	mAP@50	map@50:0.95
Baseline	0.764	0.697	0.729	0.779	0.63
Baseline + ClusterNMS	0.729	0.755	0.742	0.788	0.621

- Recall ↑
- F1-Score ↑
- Precision ↓

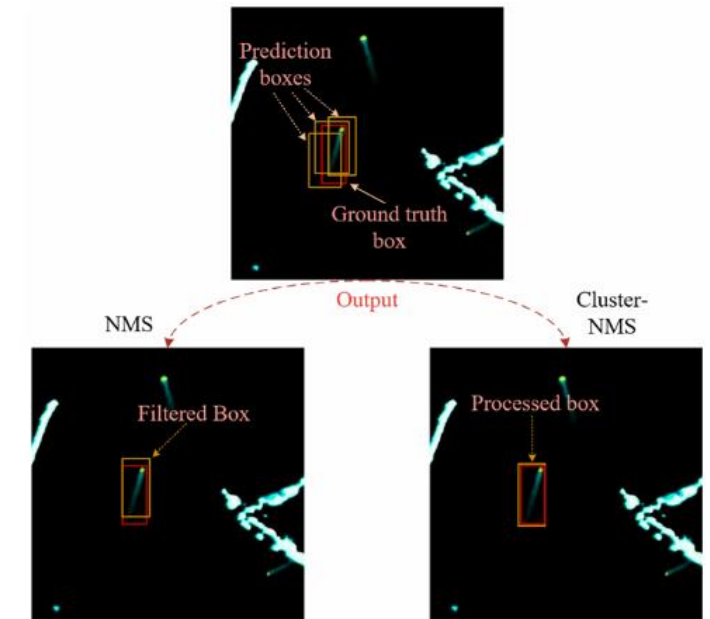
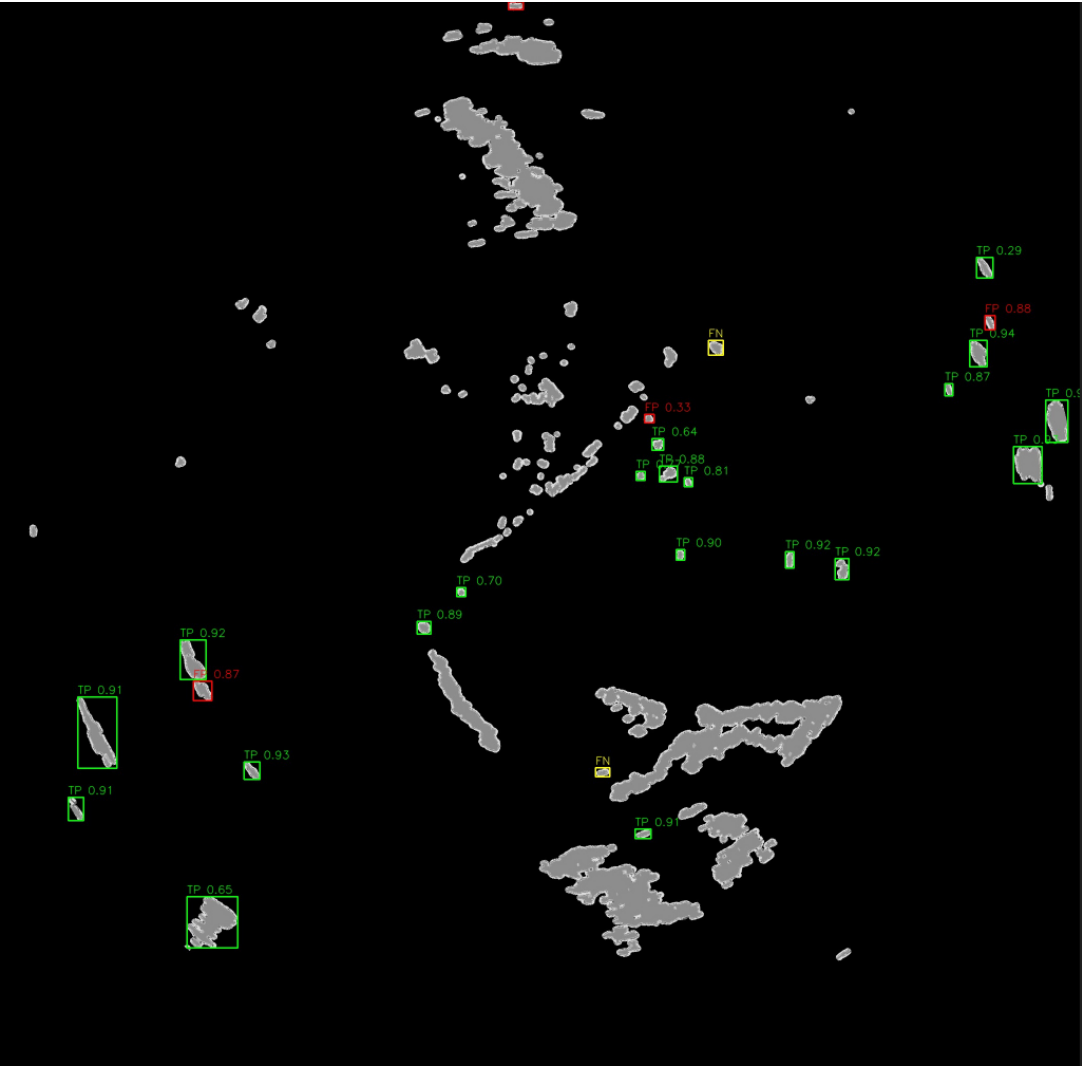
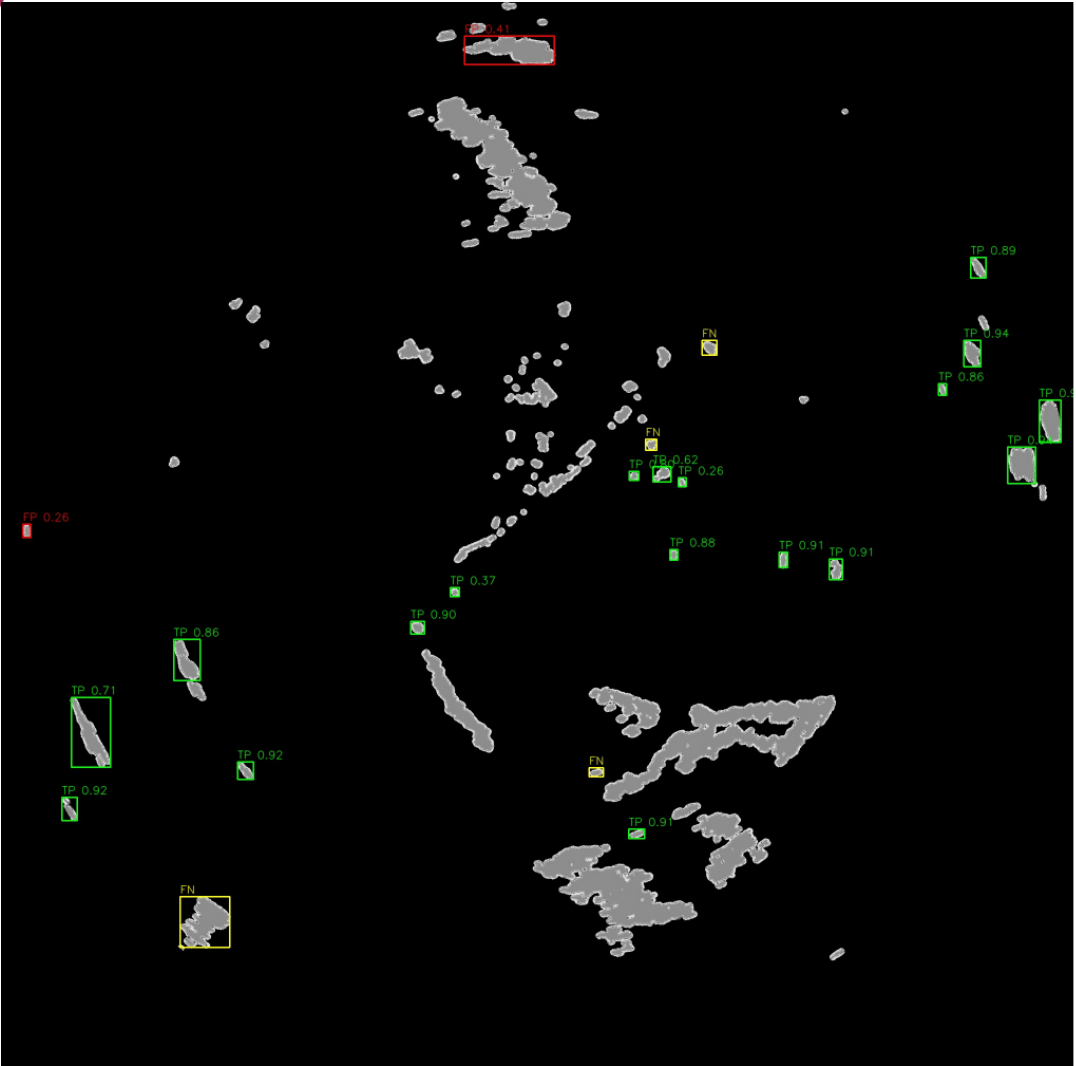
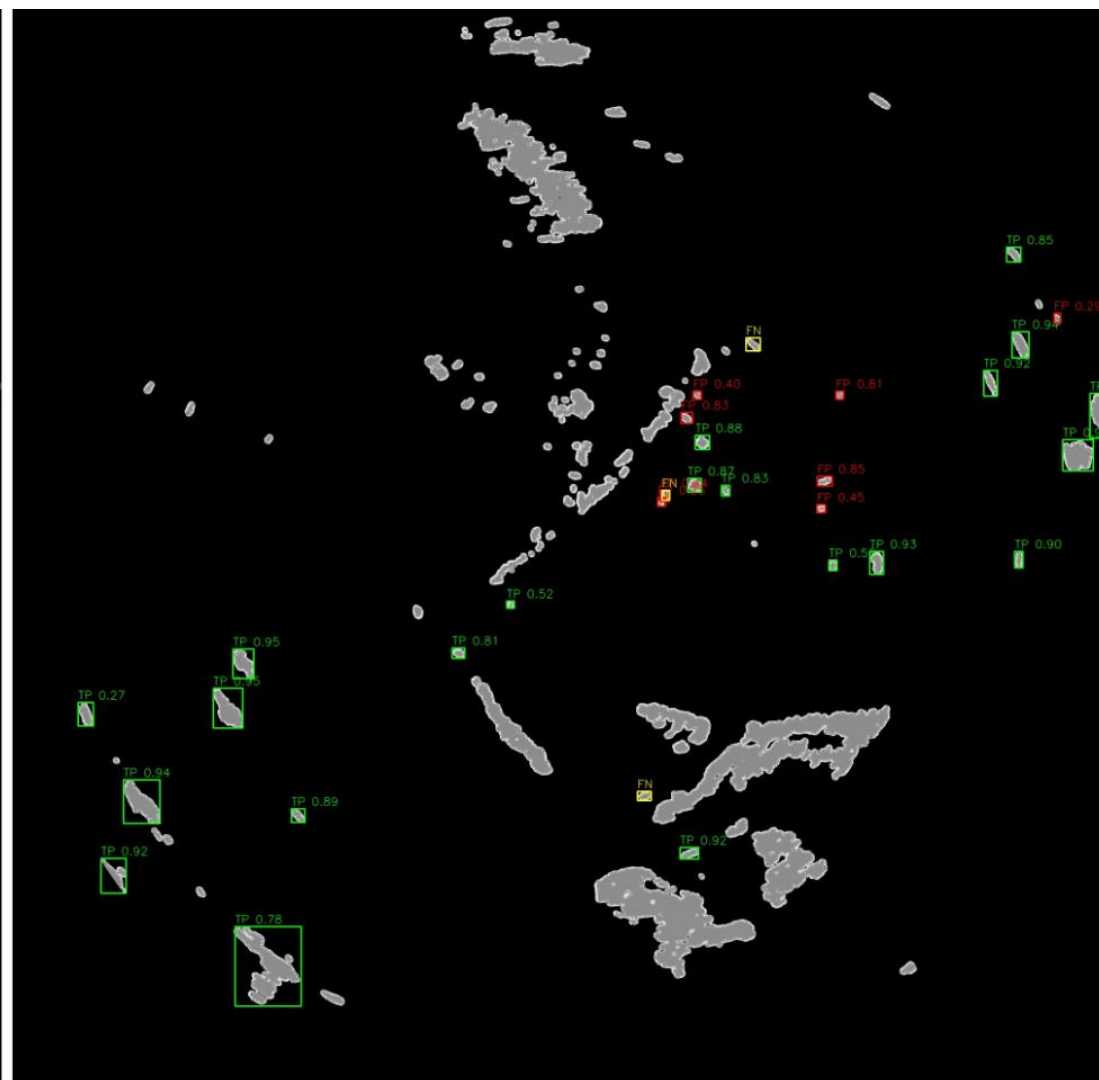
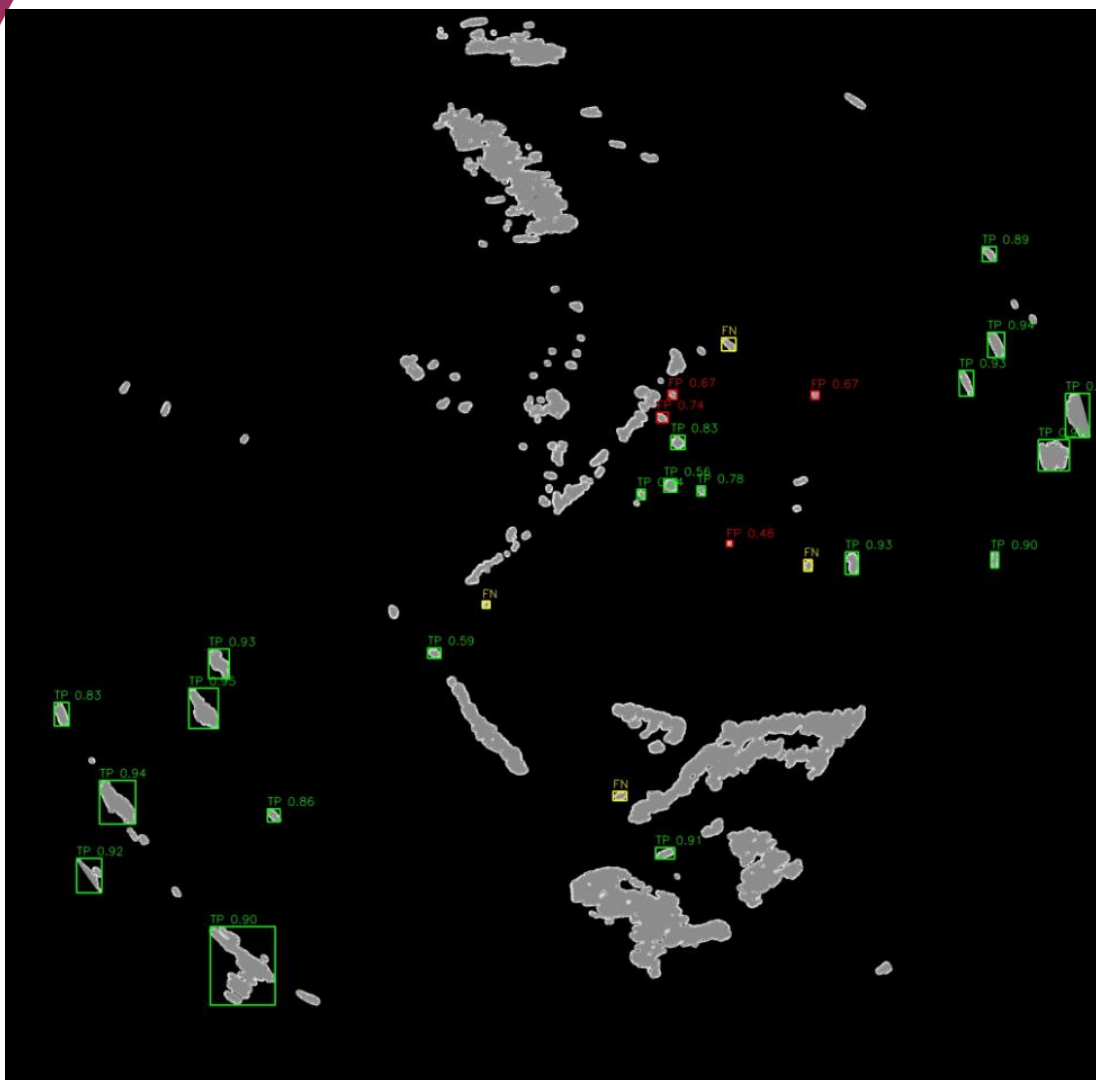


Fig 4: Comparison of NMS and Cluster-NMS — NMS removes overlapping boxes, while Cluster-NMS merges them into a single refined box, reducing missed detections and improving recall [1].

Baseline vs Baseline + ClusterNMS



Baseline vs Baseline + ClusterNMS



Baseline + α -DIoU Loss

- Standard IoU:
$$IoU = \frac{S^{pr} \cap S^{gt}}{S^{pr} \cup S^{gt}}$$

- Adding Distance IoU
$$DIoU = IoU - \frac{\rho^2}{c^2}$$

- α -DIoU Loss:
$$Loss_{\alpha-DIoU} = 1 - IoU^{2\alpha} + \left(\frac{\rho}{c}\right)^\alpha$$

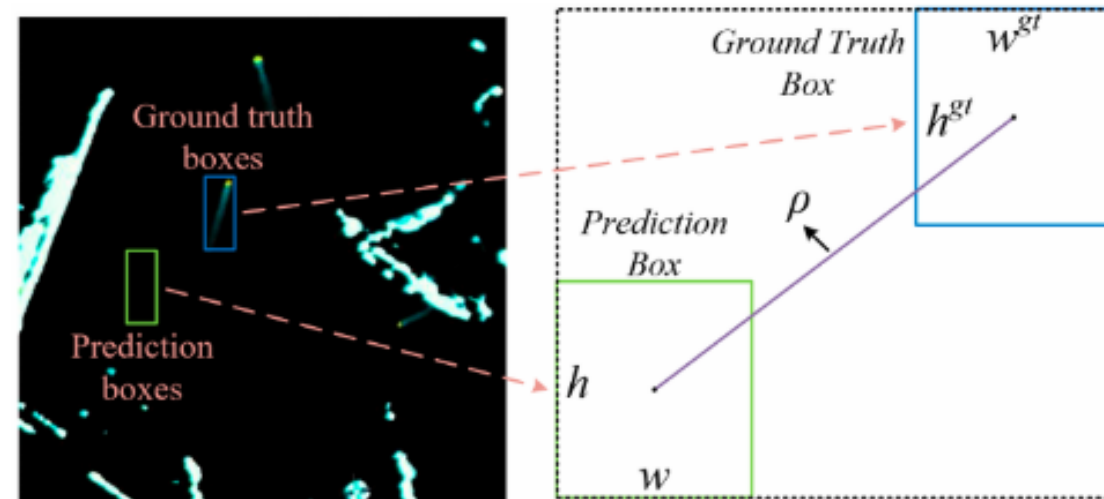


Fig 5: α -DIoU adds center distance (ρ/c) to IoU, improving box localization for small ships [1].

	Precision	Recall	F1-Score	mAP@50	map@50:0.95
Baseline	0.764	0.697	0.729	0.779	0.63
Baseline + α -DIoU Loss	0.747	0.751	0.749	0.806	0.636

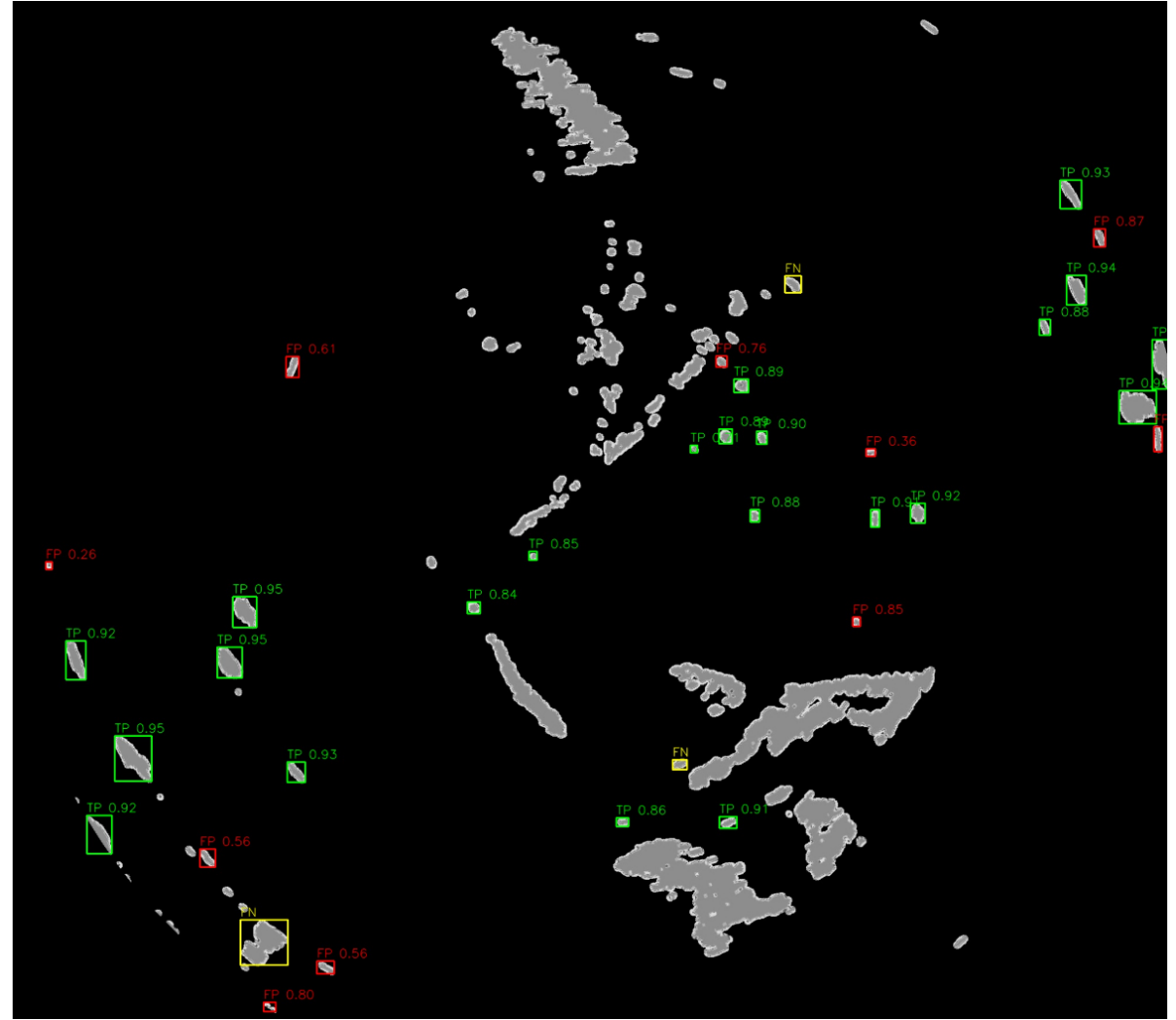
- Recall $\uparrow$
- F1-Score $\uparrow$
- Precision $\downarrow$

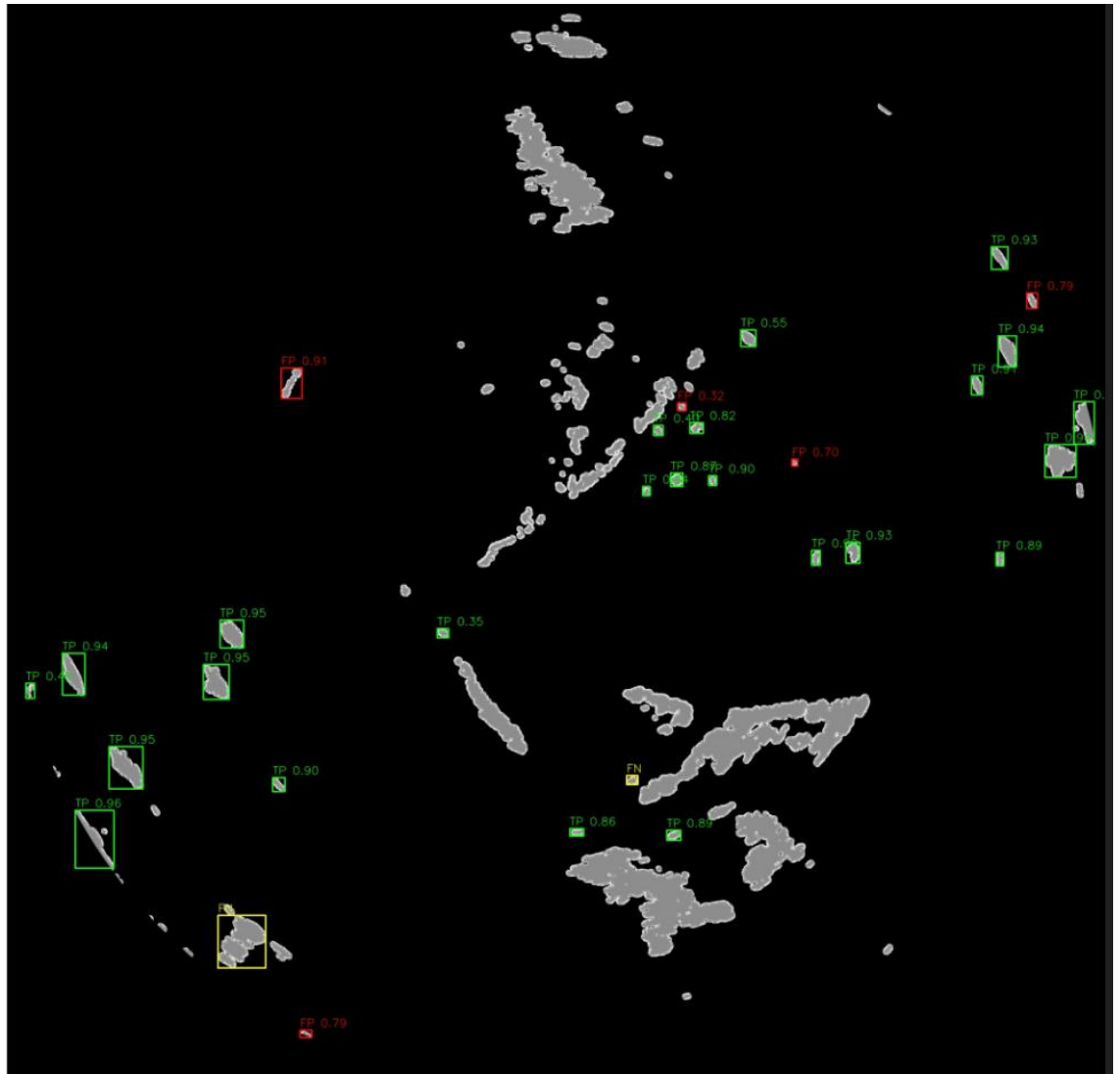
Baseline + Single Stream

- Baseline: 1 grayscale frame $\rightarrow$ 3-channel grayscale [6].
- Single-Stream:
 - Input = 3 consecutive grayscale frames, sampled with a stride of 2 (frames $t-2$, t , $t+2$). [6]
 - Short motion history $\rightarrow$ ships moving across frames become more visible [6].
 - Detection $\rightarrow$ on the middle frame [6].

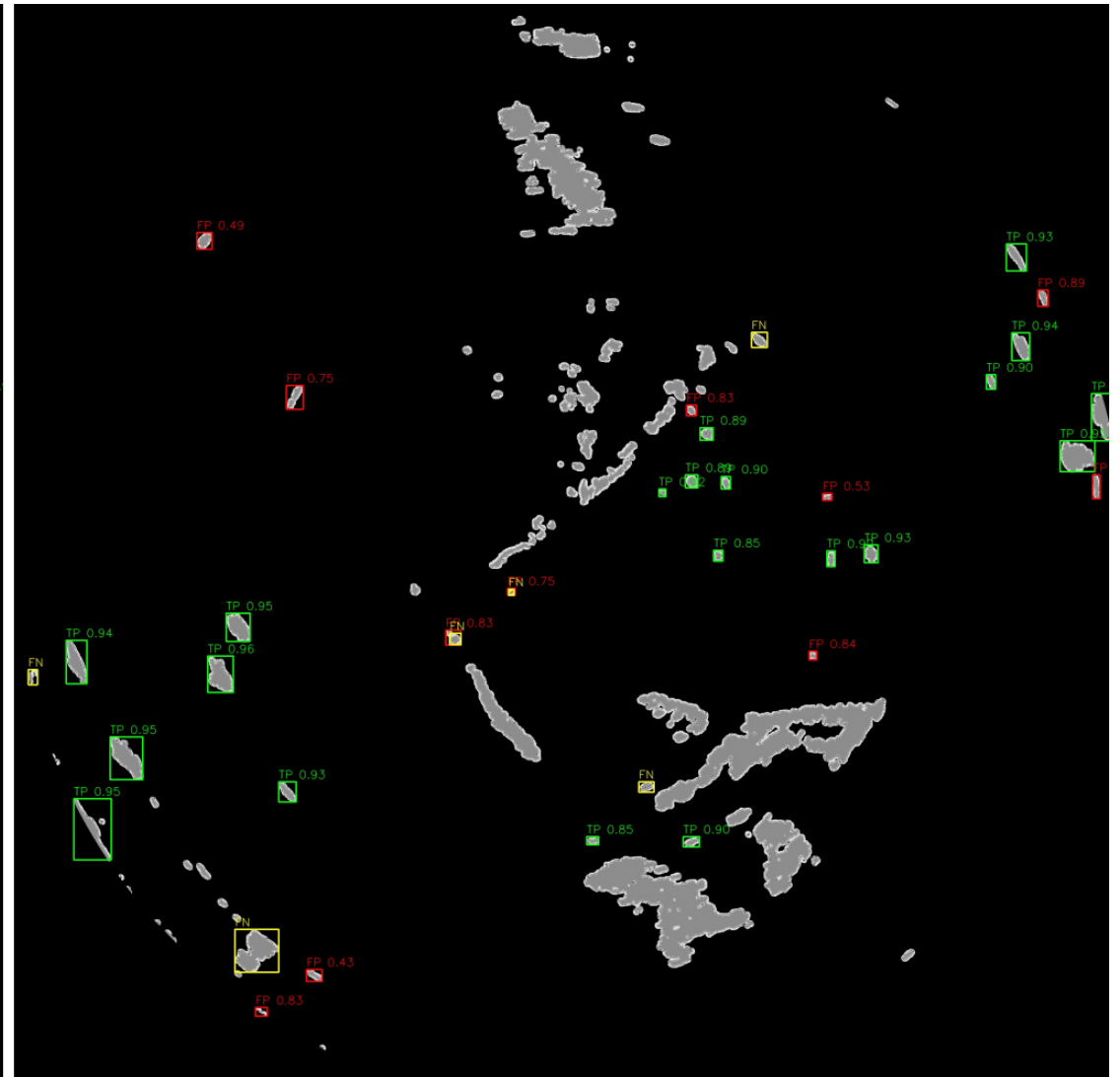
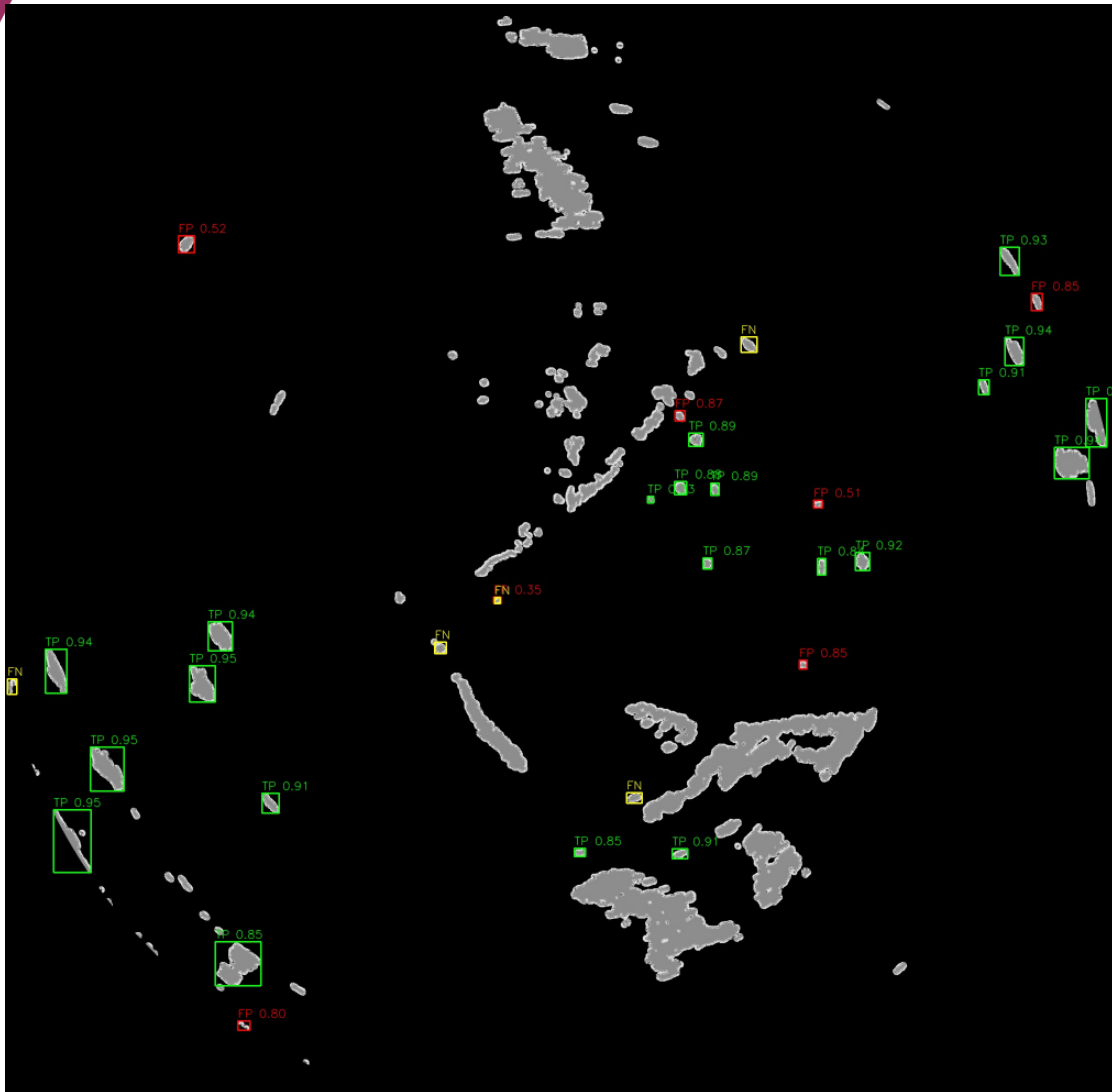
	Precision	Recall	F1-Score	mAP@50	map@50:0.95
Baseline	0.764	0.697	0.729	0.779	0.63
Baseline + Single Stream	0.711	0.724	0.717	0.777	0.609

- Recall $\uparrow$
- F1-Score $\downarrow$
- Precision $\downarrow$





Baseline vs Baseline + Single Stream



Baseline + Mobilenetv3-large Backbone

- Lightweight CNN backbone → balances accuracy and efficiency [7].
- Suitable for radar hardware with limited compute [1].
- Depthwise separable convolutions → reduce parameters & FLOPs [7].
- Squeeze-and-Excitation (SE) module → channel attention [7].
- Dilated + pointwise convolutions → multi-scale receptive fields [7].

	Precision	Recall	F1-Score	mAP@50	map@50:0.95
Baseline	0.764	0.697	0.729	0.779	0.63
Baseline + Mobilenet v3-l	0.79	0.678	0.73	0.774	0.57

- Recall ↓
- F1-Score →
- Precision ↑

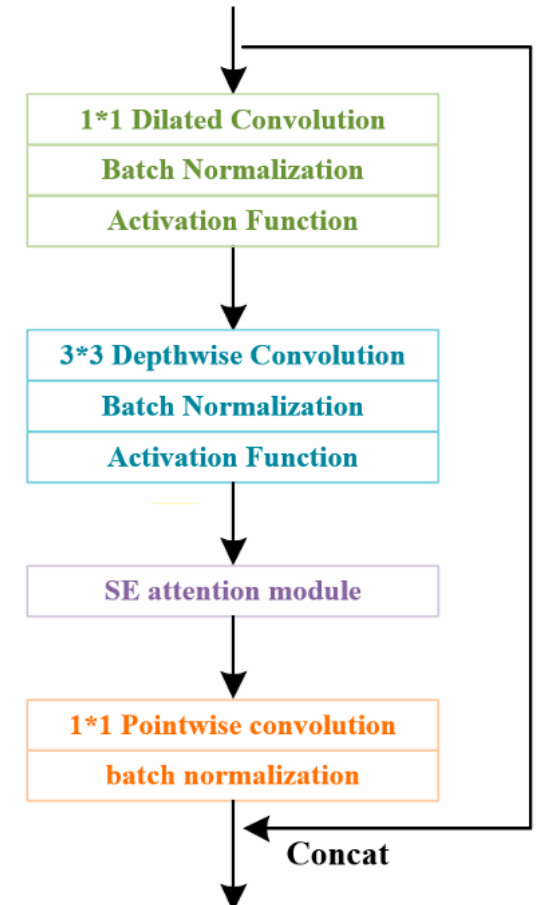
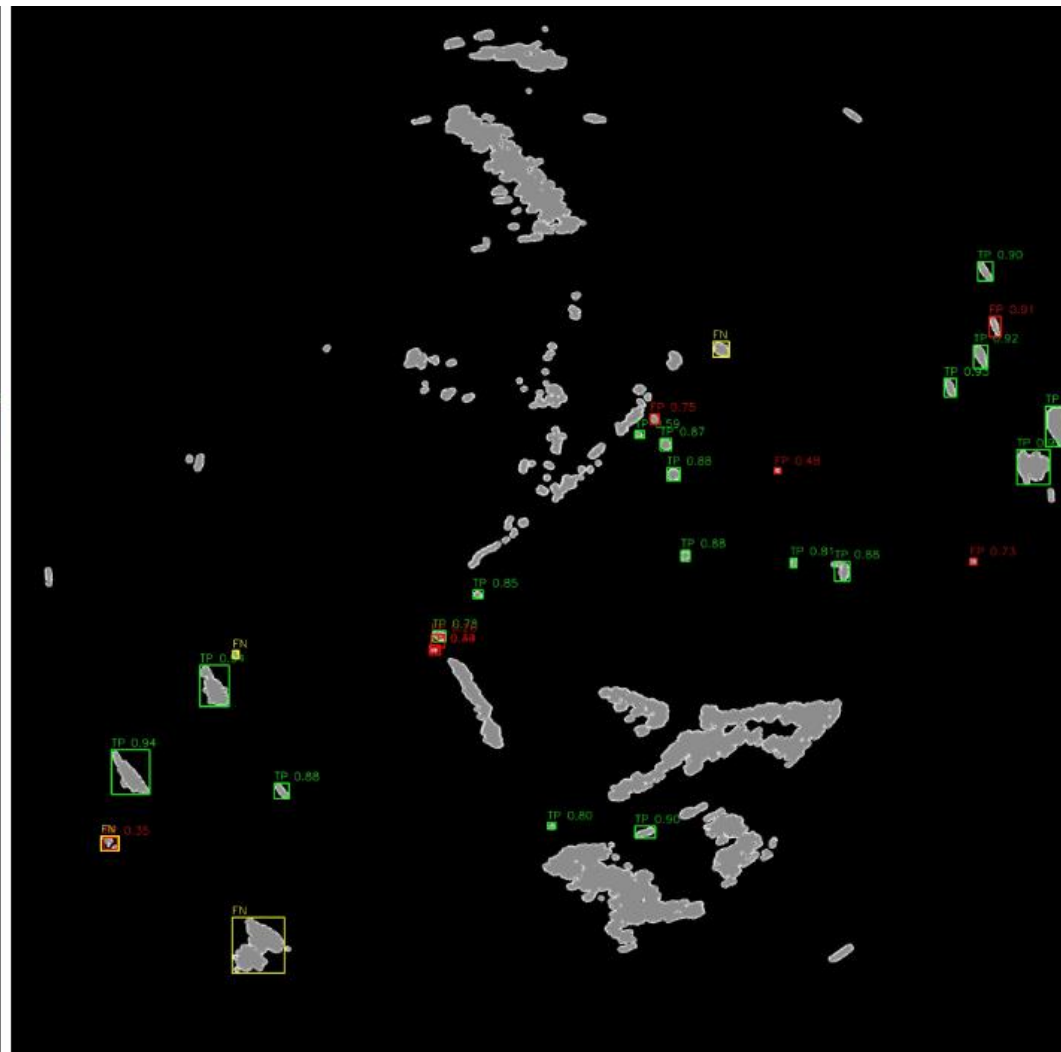
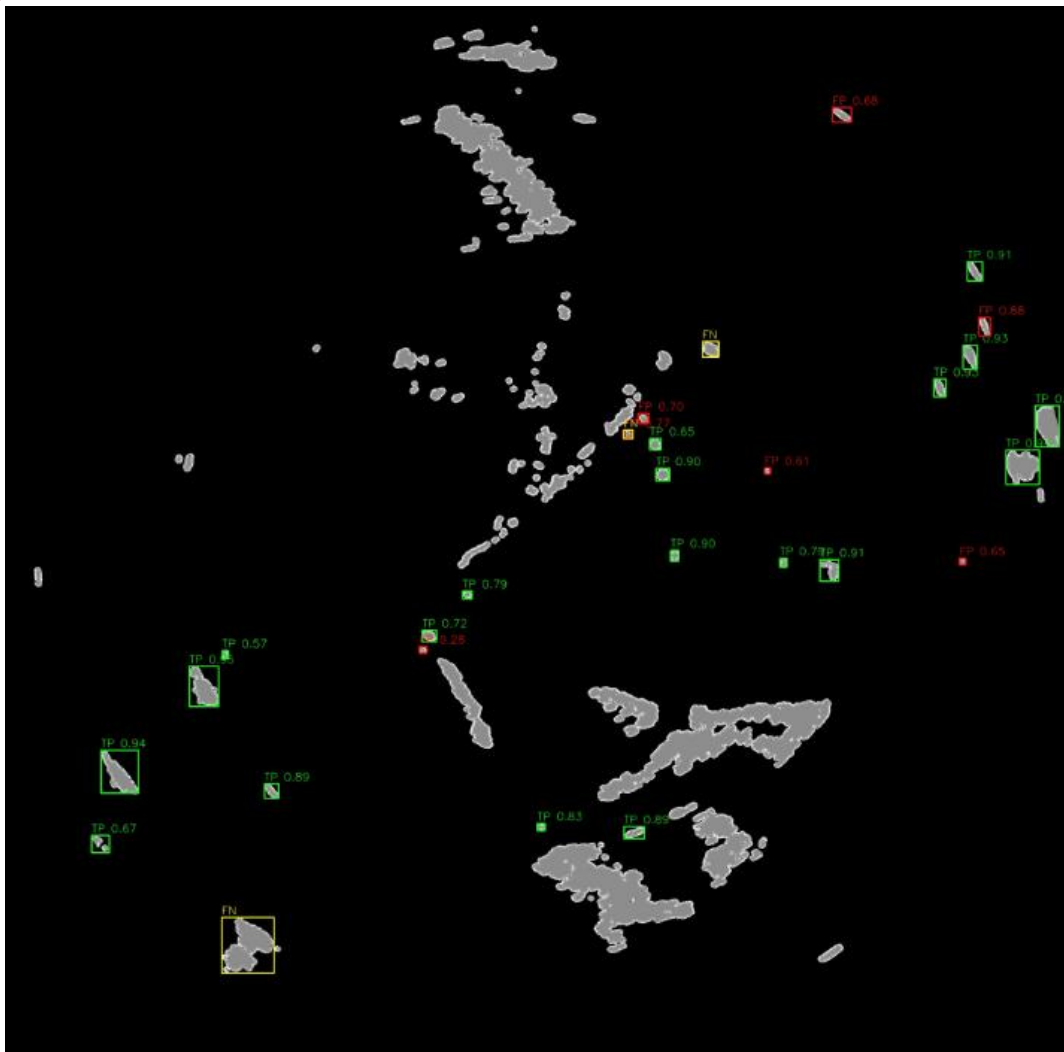
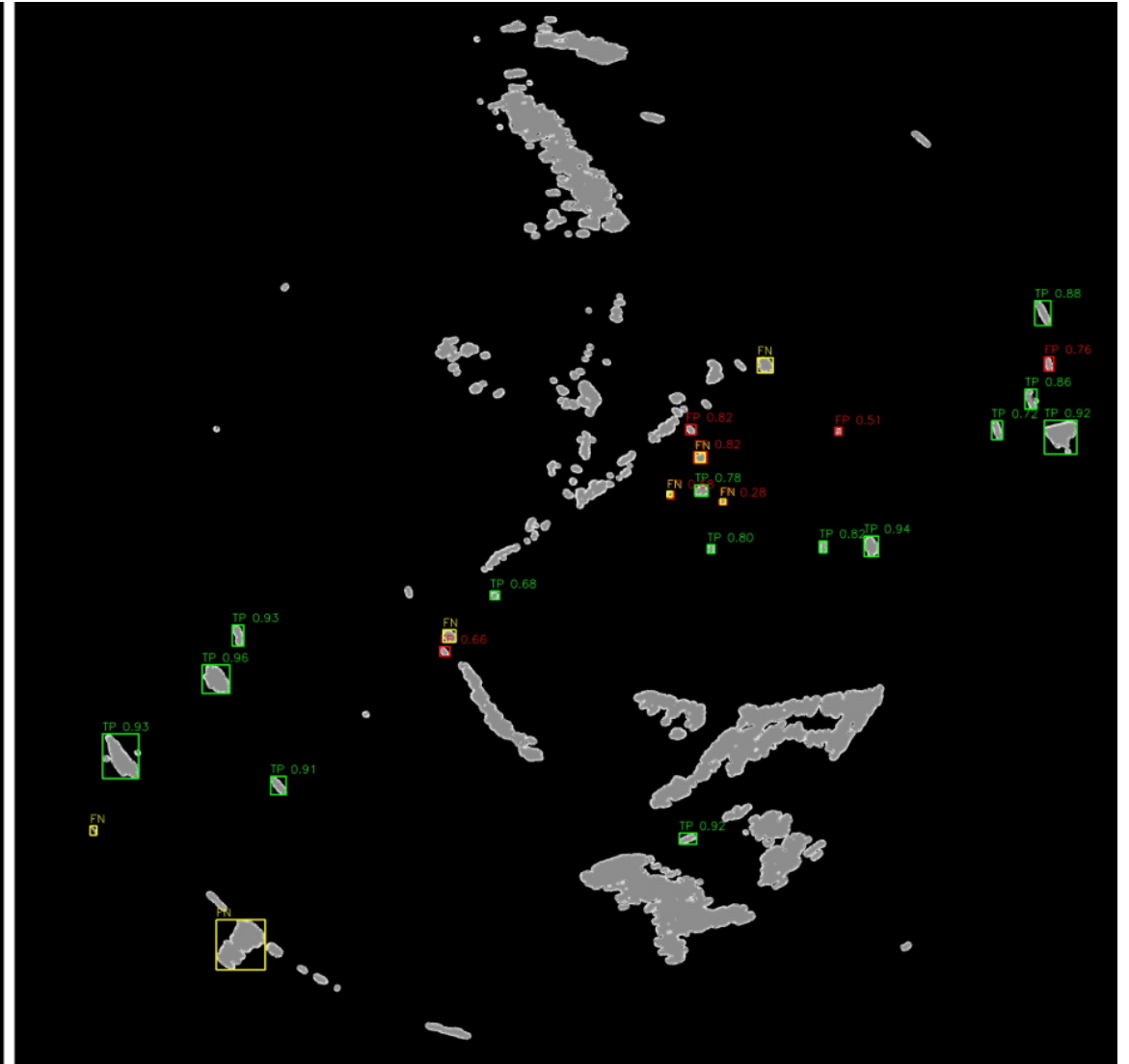
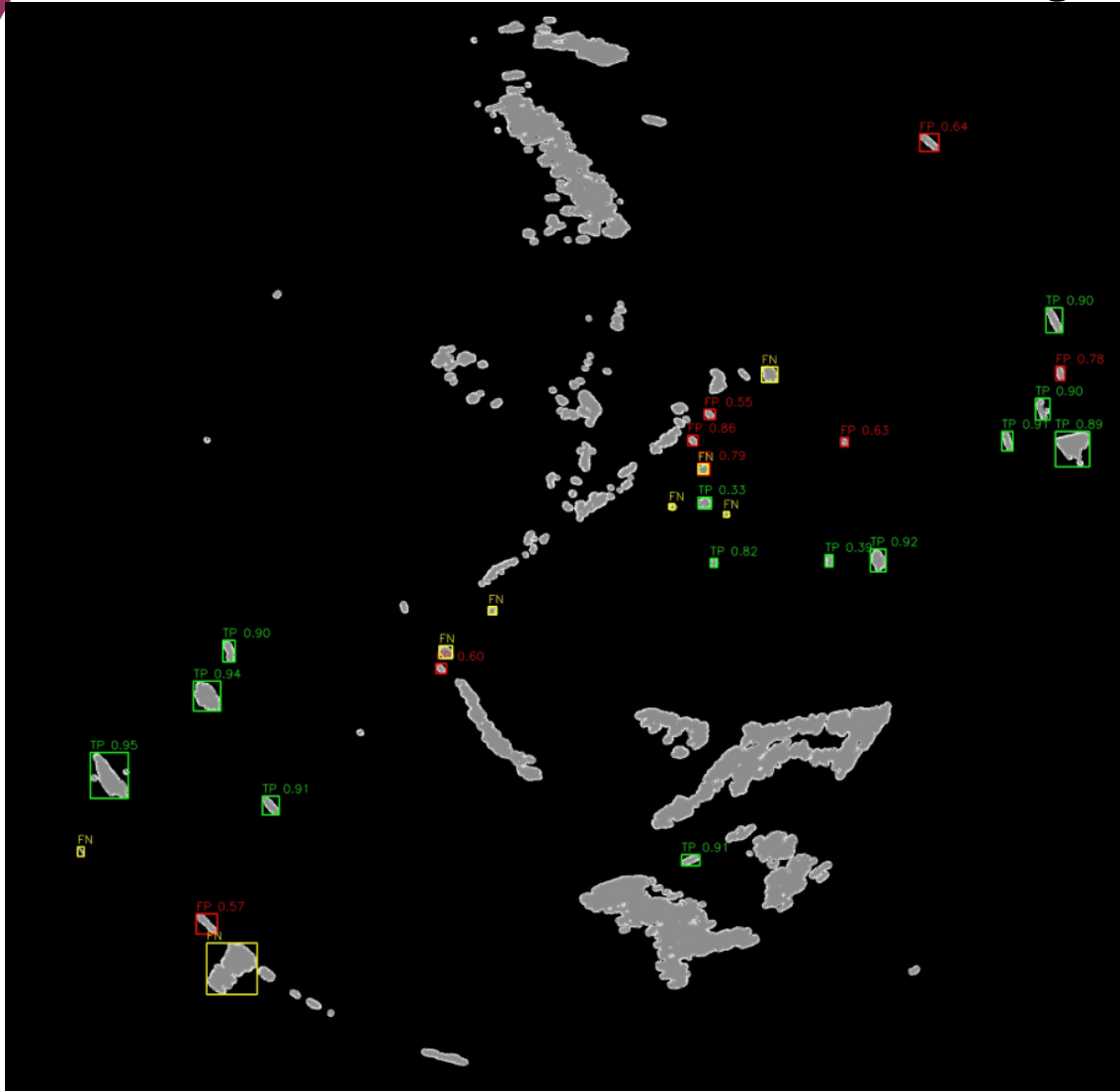


Fig 6: MobileNetV3 block structure with dilated, depthwise, SE, and pointwise convolutions for efficient multi-scale feature extraction [1].

Baseline vs Baseline + Mobilenetv3-large



Baseline vs Baseline + Mobilenetv3-large



Baseline + SPP YOSMR

- Multi-scale pooling kernels $\rightarrow$ local / mid-level / global context [1].
- Capture objects at different scales $\rightarrow$ small ships & large ships [1].
- YOLOv5 $\rightarrow$ includes fast version (SPPF) [5].
 - SPPF (YOLOv5): progressive pooling (5, 9, 13), lightweight
 - SPP (YOSMR): 4 independent branches (1, 5, 9, 13), richer multi-scale details

	Precision	Recall	F1-Score	mAP@50	map@50: 0.95
Baseline	0.764	0.697	0.729	0.779	0.63
Baseline + SPP YOSMR	0.767	0.709	0.737	0.777	0.624

- Recall $\uparrow$
- F1-Score $\uparrow$
- Precision $\sim$

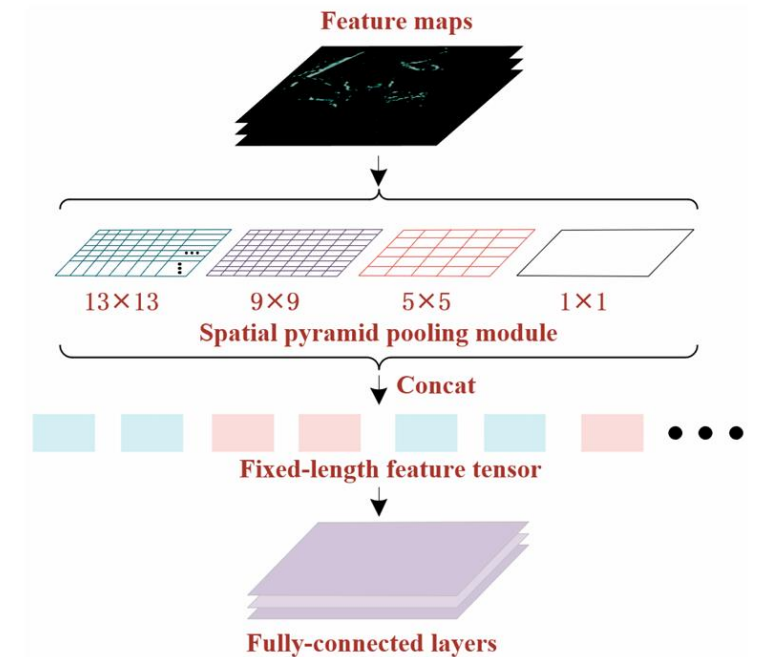
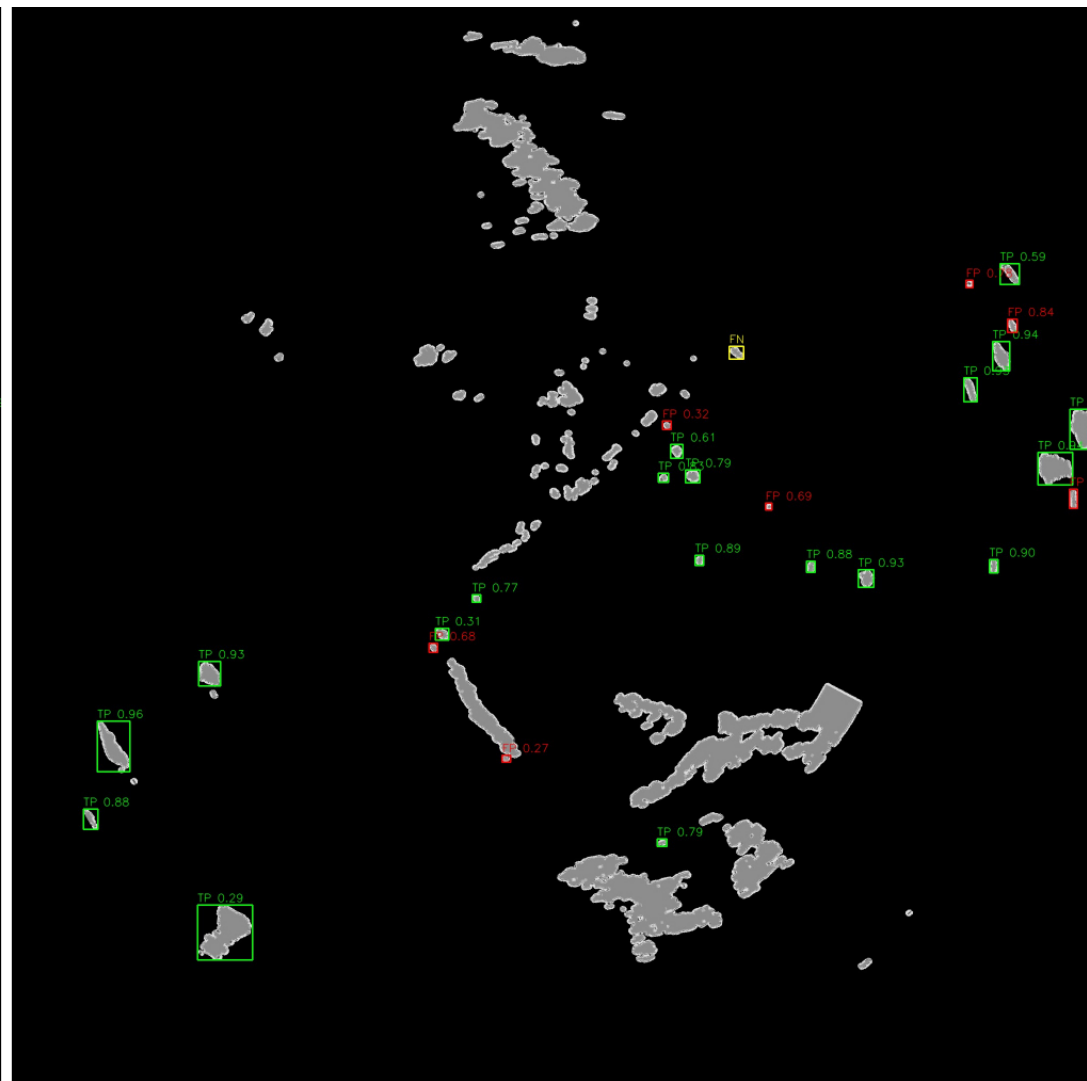
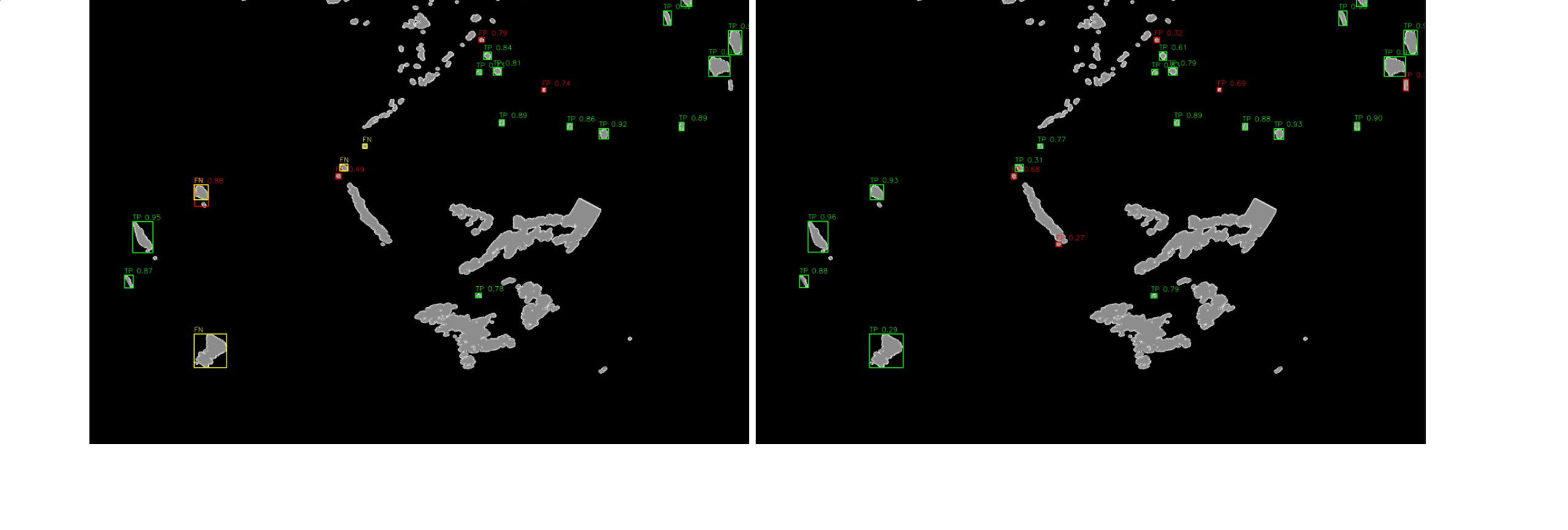


Fig 7: Spatial Pyramid Pooling (SPP) uses multiple kernel sizes to capture local and global features, then concatenates them into a fixed-length feature tensor [1].

Baseline vs Baseline + SPP YOSMR



Baseline + LightPANet

- YOLOv5 head → PANet for feature fusion [5].
- LightPANet [1].
 - C3 → C3Light
 - Conv → DSConv

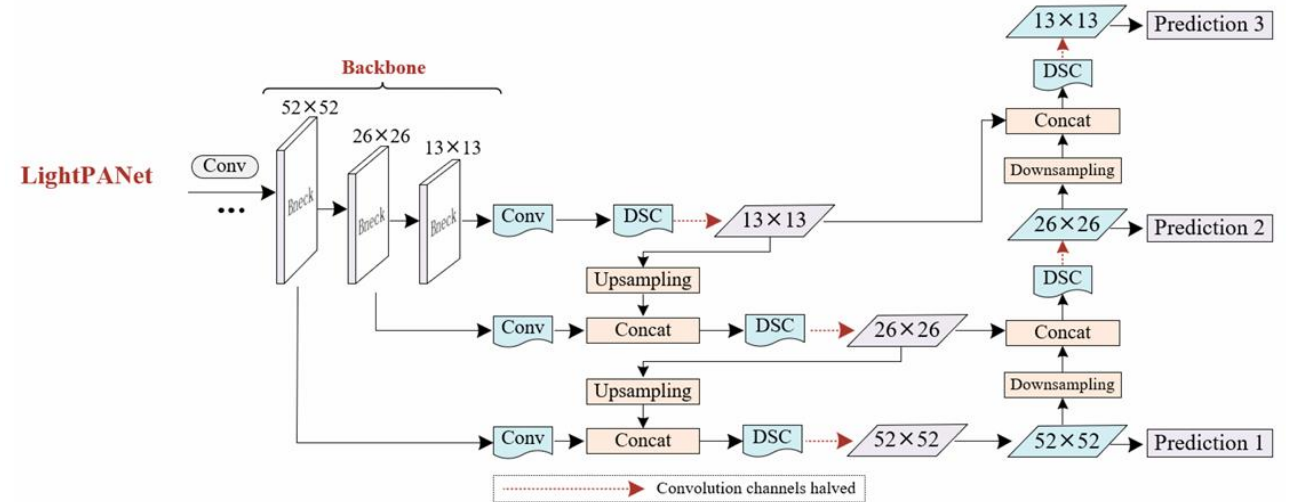
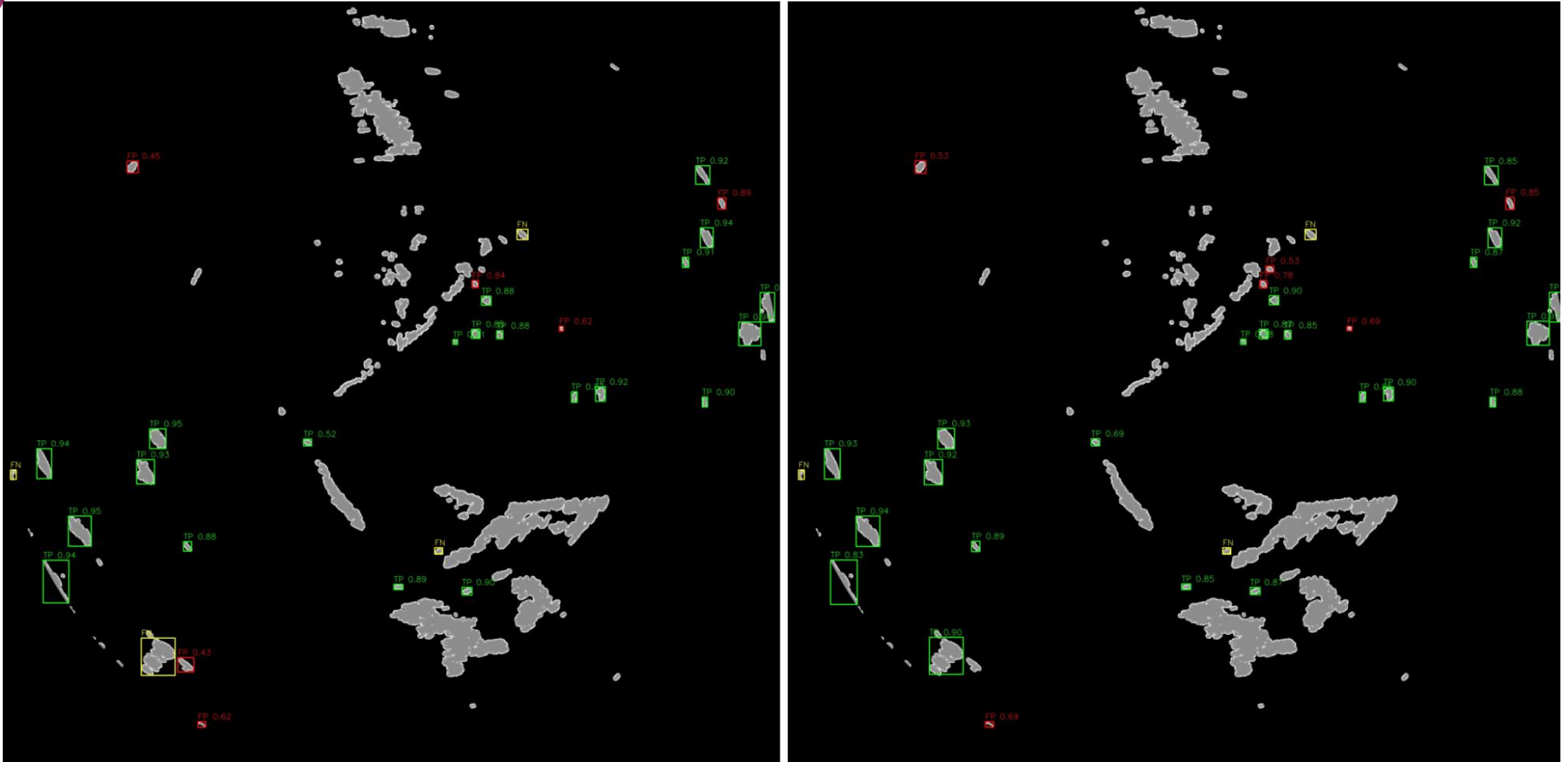


Fig 8: LightPANet structure showing feature fusion with depthwise separable convolutions and lightweight C3Light blocks across multiple scales [1].

	Precision	Recall	F1-Score	mAP@50	map@50:0.95
Baseline	0.764	0.697	0.729	0.779	0.63
Baseline + LightPANet	0.76	0.701	0.729	0.779	0.6

Recall ~ ↑
 F1-Score ~
 Precision ~

Baseline vs Baseline + LightPANet



Baseline + Radar Specific Augmentations

- **Baseline Augmentations**

- **Geometric:**

- Rotation: $\pm 2^\circ$, Translate: 10%, Scale: 0.80, Shear: 0.10
- Flip LR: 50% (No vertical flip, No perspective)

- **Mixing:**

- Mosaic: 70%, MixUp: 5%, Copy-Paste: 0

- **Color:** Disabled (grayscale radar)

- **Radar-specific:** None

- ▶ **Enhanced Augmentations**

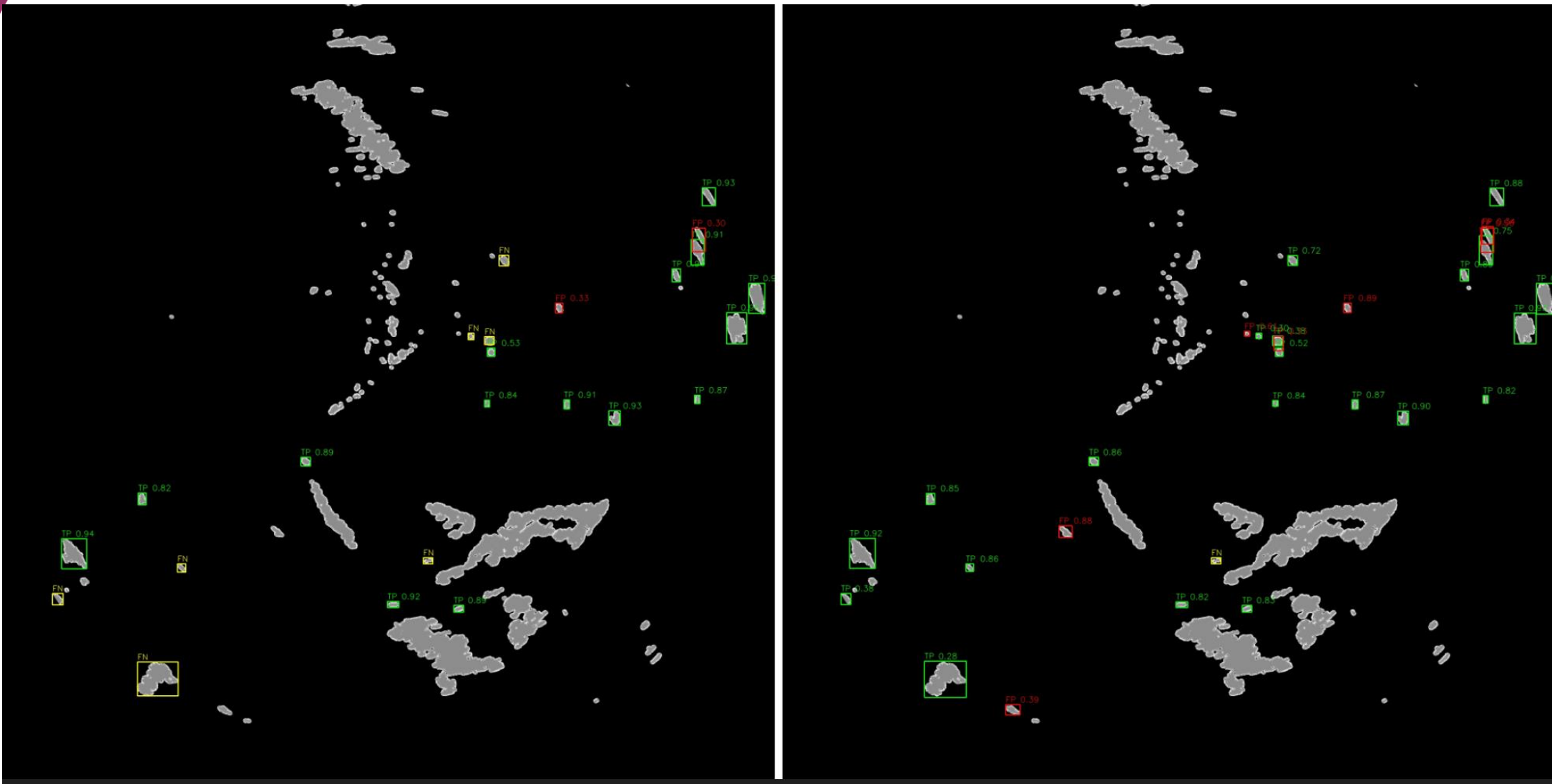
- ▶ **Radar-specific (new):**

- ▶ Noise injection (Gaussian + speckle, 5%)
- ▶ Blur (10%)
- ▶ Cutout (10%)
- ▶ RCS scaling (0.8–1.2, 30%)
- ▶ Azimuth blur (20%, kernel=5)
- ▶ Ghost echoes (10%)
- ▶ Range distortion (20%)

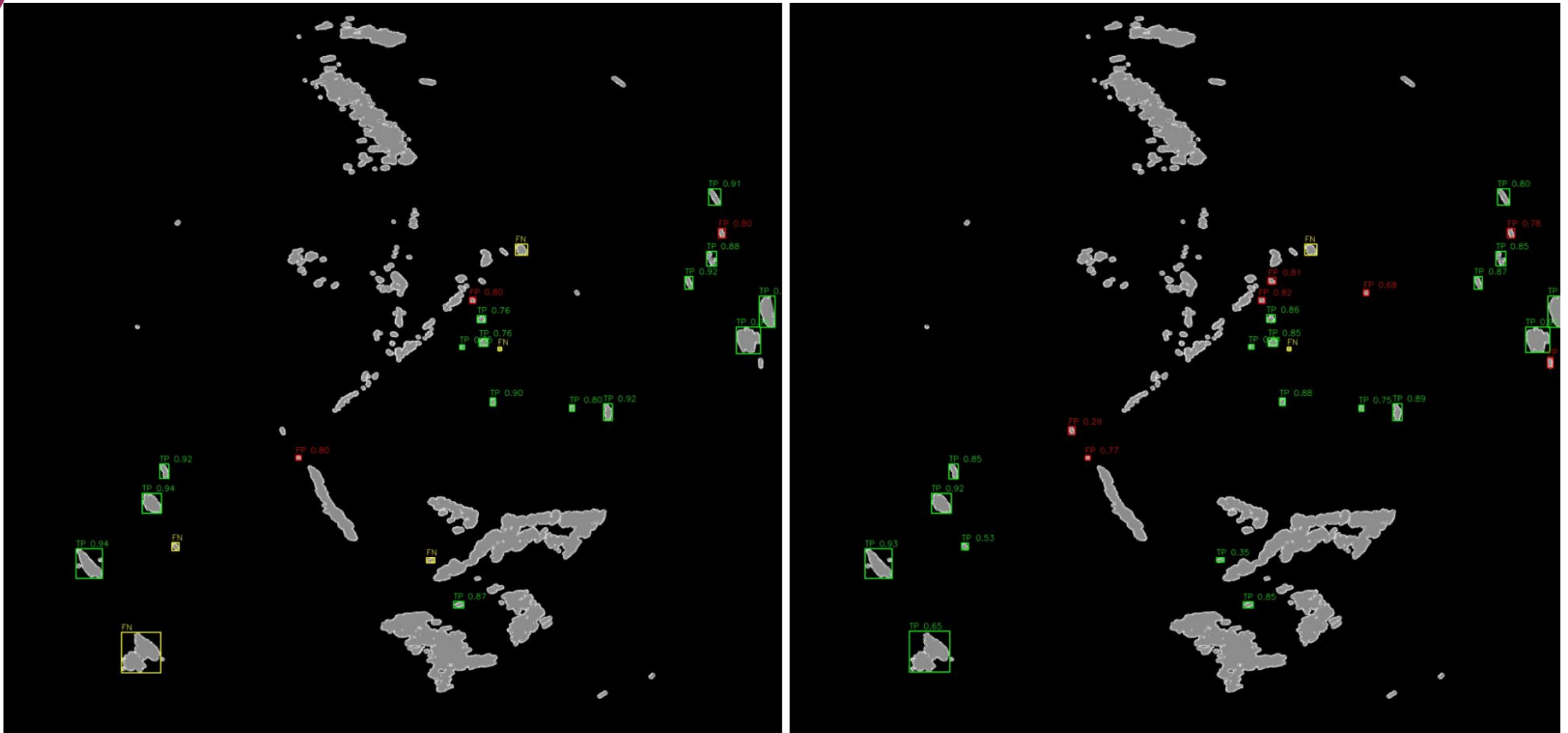
	Precision	Recall	F1-Score	mAP@50	map@50:0.95
Baseline	0.764	0.697	0.729	0.779	0.63
Baseline + Radar Specific Augmentations	0.762	0.725	0.743	0.811	0.598

- Recall ↑
- F1-Score ↑
- Precision ~

Baseline vs Baseline + Radar Specific Augmentations



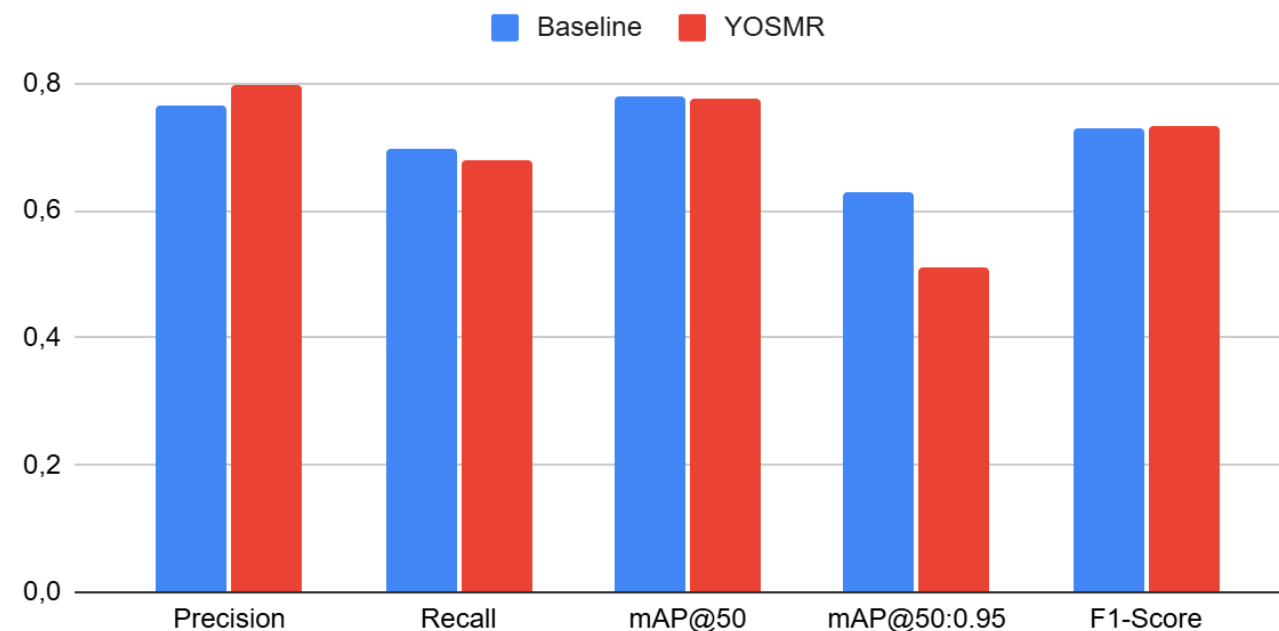
Baseline vs Baseline + Radar Specific Augmentations



Baseline + All Modules Combined → YOSMR

- ClusterNMS
- α -DIoU Loss
- Mobilenetv3-Large
- SPP
- LightPANet
- Augmentations

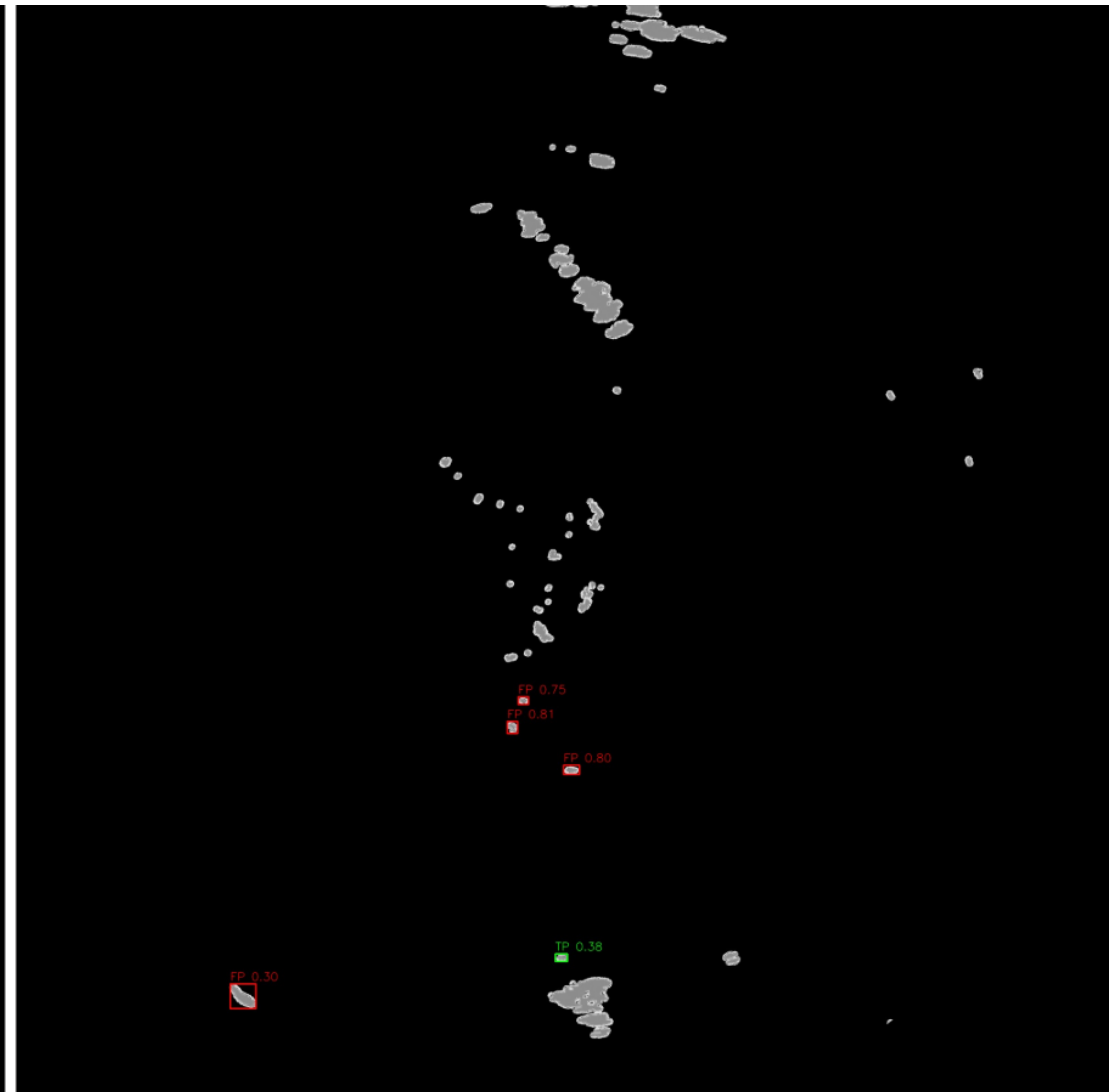
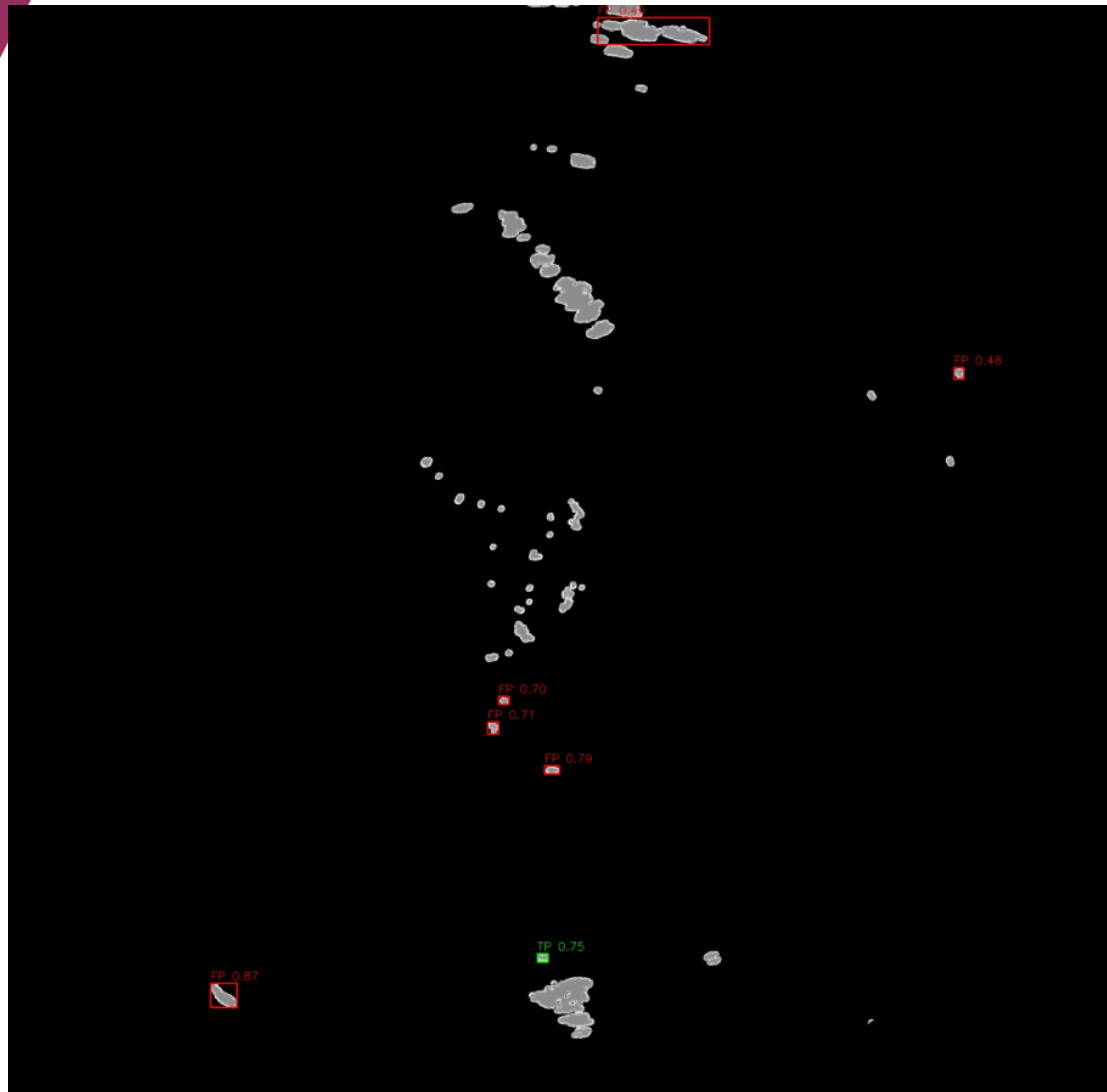
Baseline vs YOSMR



	Precision	Recall	F1-Score	mAP@50	Experiment map@50:0.9 5
Baseline	0.764	0.697	0.729	0.779	0.63
YOSMR	0.798	0.678	0.733	0.776	0.509

- Recall ↓
- F1-Score ↑
- Precision ↑

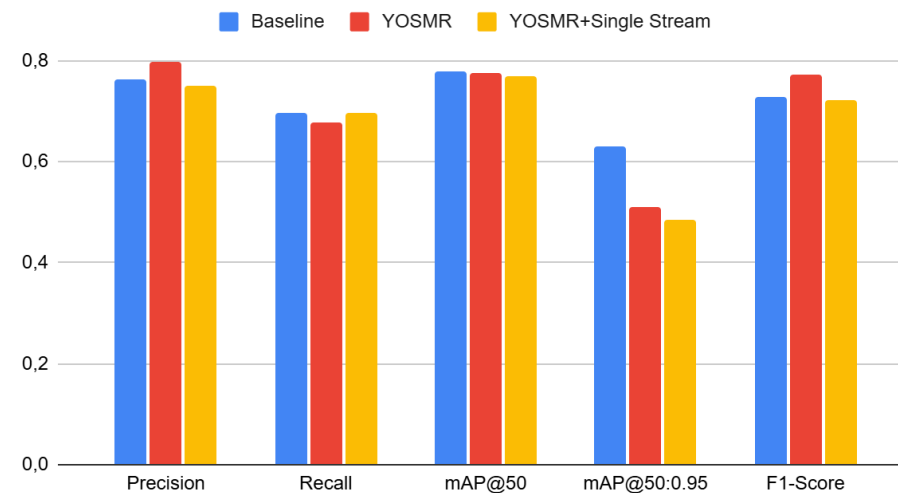
Baseline vs YOSMR



Baseline + All Modules + Single Stream

	Baseline	YOSMR	YOSMR + Single Stream
Precision	0,764	0,798	0,75
Recall	0,697	0,678	0,696
mAP@50	0,779	0,776	0,77
mAP@50:0.95	0,63	0,509	0,484
F1-Score	0,729	0,733	0,722

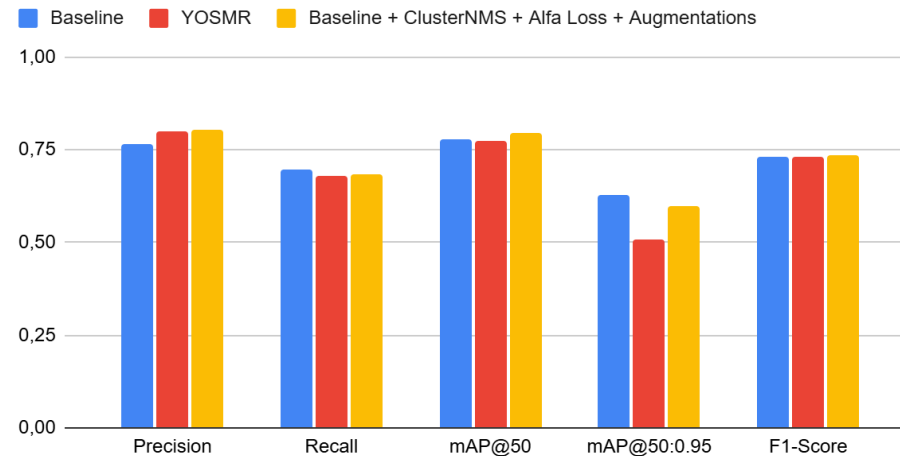
Baseline, YOSMR and YOSMR+Single Stream



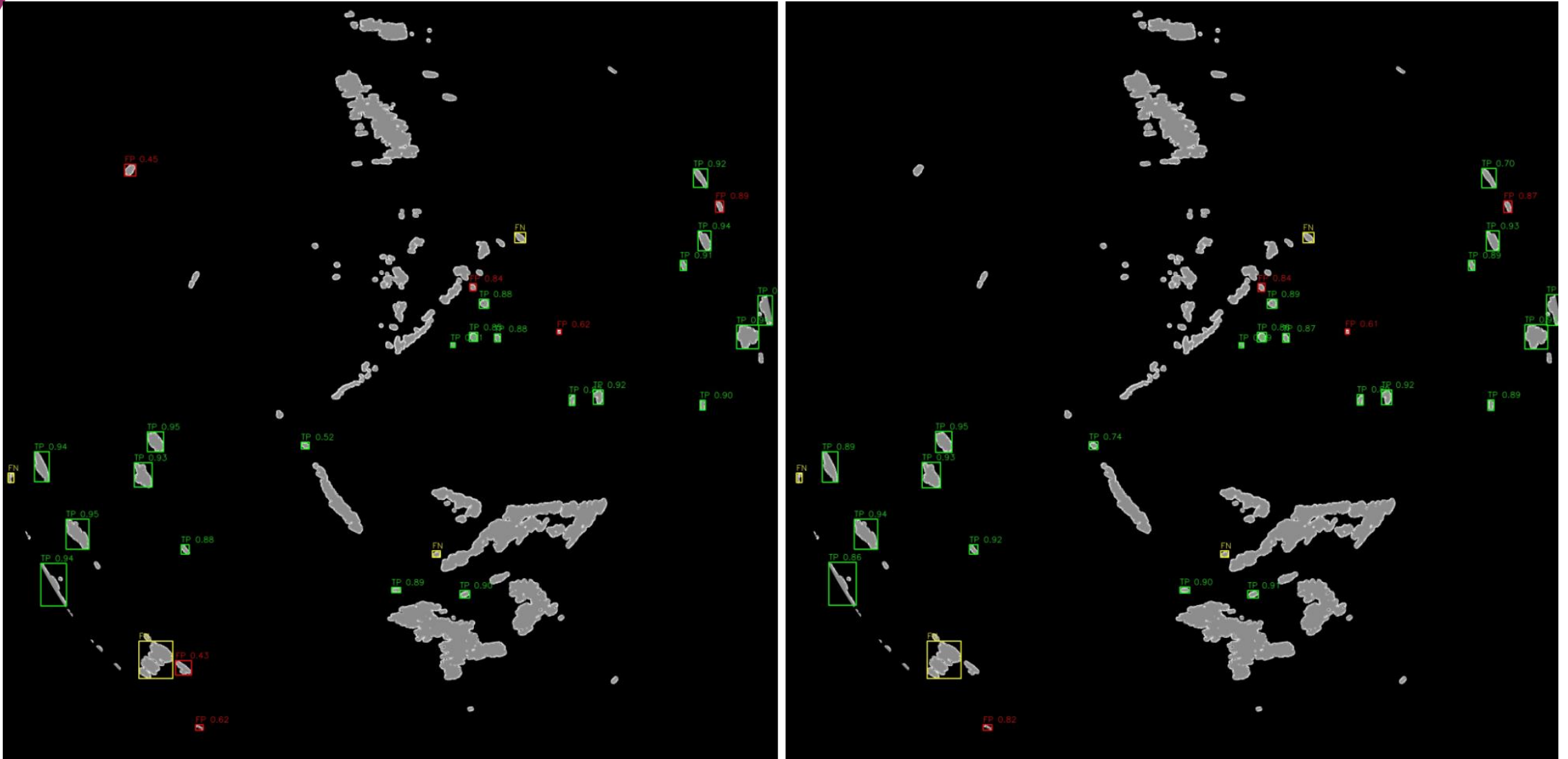
Best Model: Baseline+ ClusterNMS + α -DloU Loss + Augmentations

	Baseline	YOSMR	Best Model
Precision	0,764	0,798	0,805
Recall	0,697	0,678	0,682
mAP@50	0,779	0,776	0,796
mAP@50:0.95	0,63	0,509	0,6
F1-Score	0,729	0,733	0,735

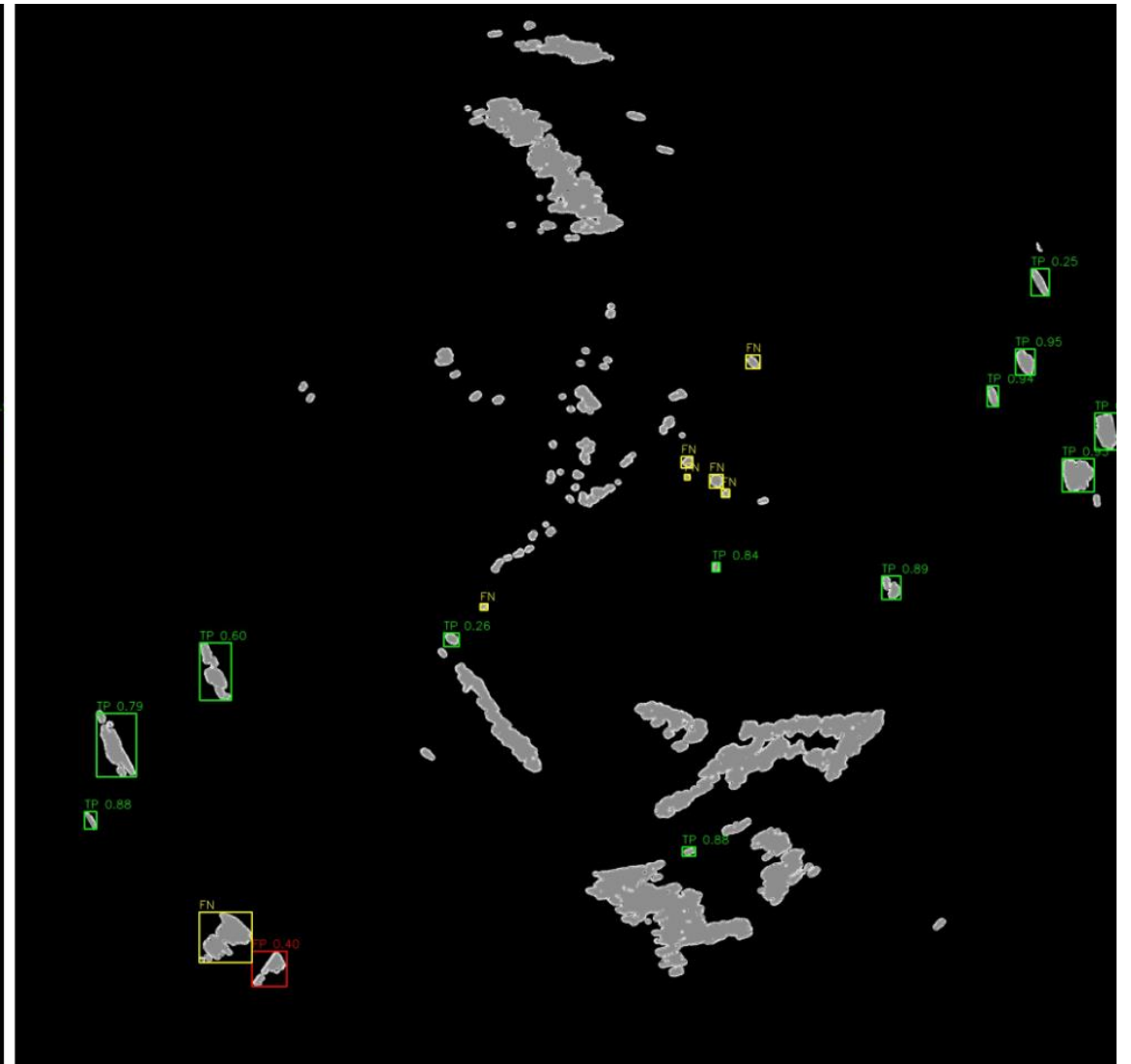
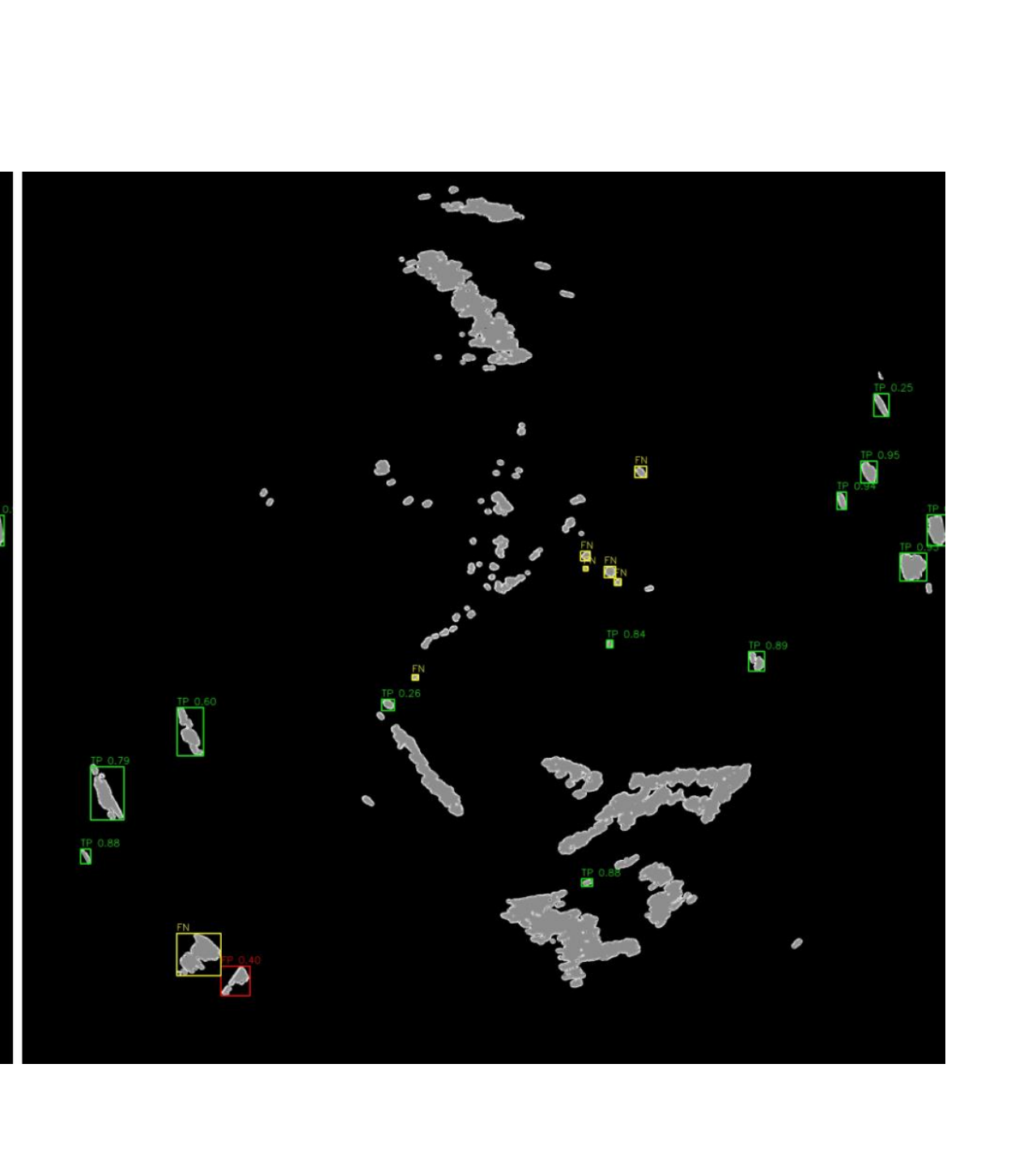
Baseline, YOSMR, and Baseline + ClusterNMS + Alfa Loss + Augmentations



Baseline vs Best Model



Baseline vs Best Model



Next Steps & Suggestions

- Dataset diversification
- Backbone improvement
- Ensemble Learning

References

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- [6] C. W. Corsel, M. van Lier, L. Kampmeijer, N. Boehrer, and E. M. Bakker, "Exploiting Temporal Context for Tiny Object Detection," in *Proc. IEEE/CVF Winter Conf. on Applications of Computer Vision (WACV) Workshops*, Waikoloa, HI, USA, Jan. 2023, pp. 640–649. doi: 10.1109/WACVW58289.2023.00074.
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Thank you for your time and attention.